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CHOI et al.(10) **Pub. No.: US 2025/0283640 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **REFRIGERATOR AND METHOD FOR
CONTROLLING THE SAME****Publication Classification**(71) Applicant: **SAMSUNG ELECTRONICS CO.,
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LTD.**, Suwon-si (KR)(21) Appl. No.: **19/096,042**(22) Filed: **Mar. 31, 2025****Related U.S. Application Data**(63) Continuation of application No. PCT/KR2025/
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Mar. 5, 2024 (KR) 10-2024-0031628

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(2013.01)

(57)

ABSTRACT

A refrigerator may include: a storage compartment; a cooling sink (CS); a thermoelectric element including a cooling layer contacting the CS, the thermoelectric element configured to, based on a voltage pulse applied to the thermoelectric element, cool the cooling layer and thereby cool the CS; a fan controllable to generate an airflow to be cooled by the CS; a cooling duct covering the CS and fan, and configured to guide the airflow from the storage compartment to the CS and to accommodate defrost water generated by water, frozen by the CS, being defrosted; a temperature sensor to detect a temperature of the CS; and a processor configured to: control the cooling fan to evaporate the defrost water, and perform an anti-freezing control so that the defrost water does not freeze, the anti-freezing control including adjusting a duty ratio of the voltage pulse based on the detected temperature of the CS.

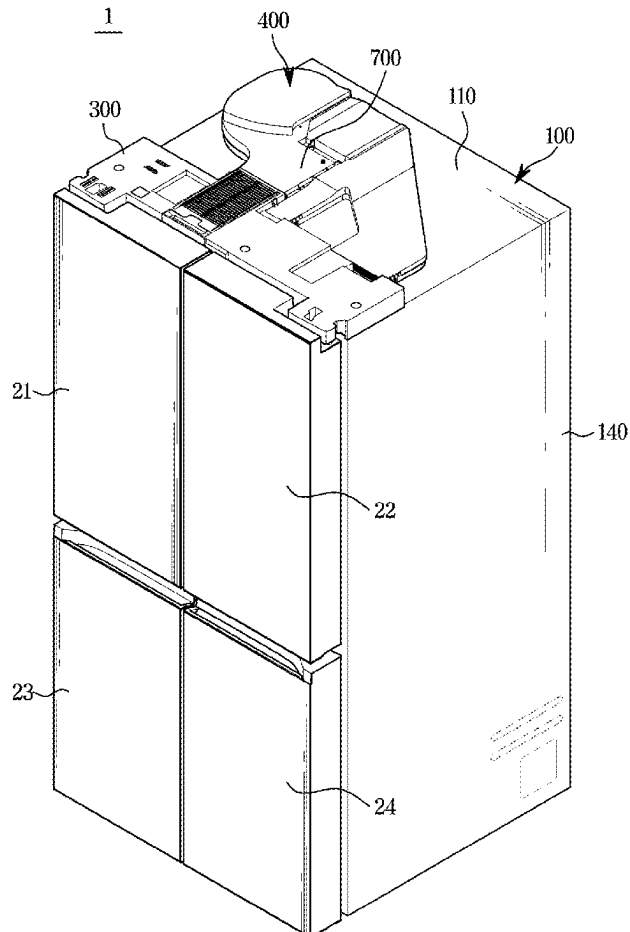


FIG. 1

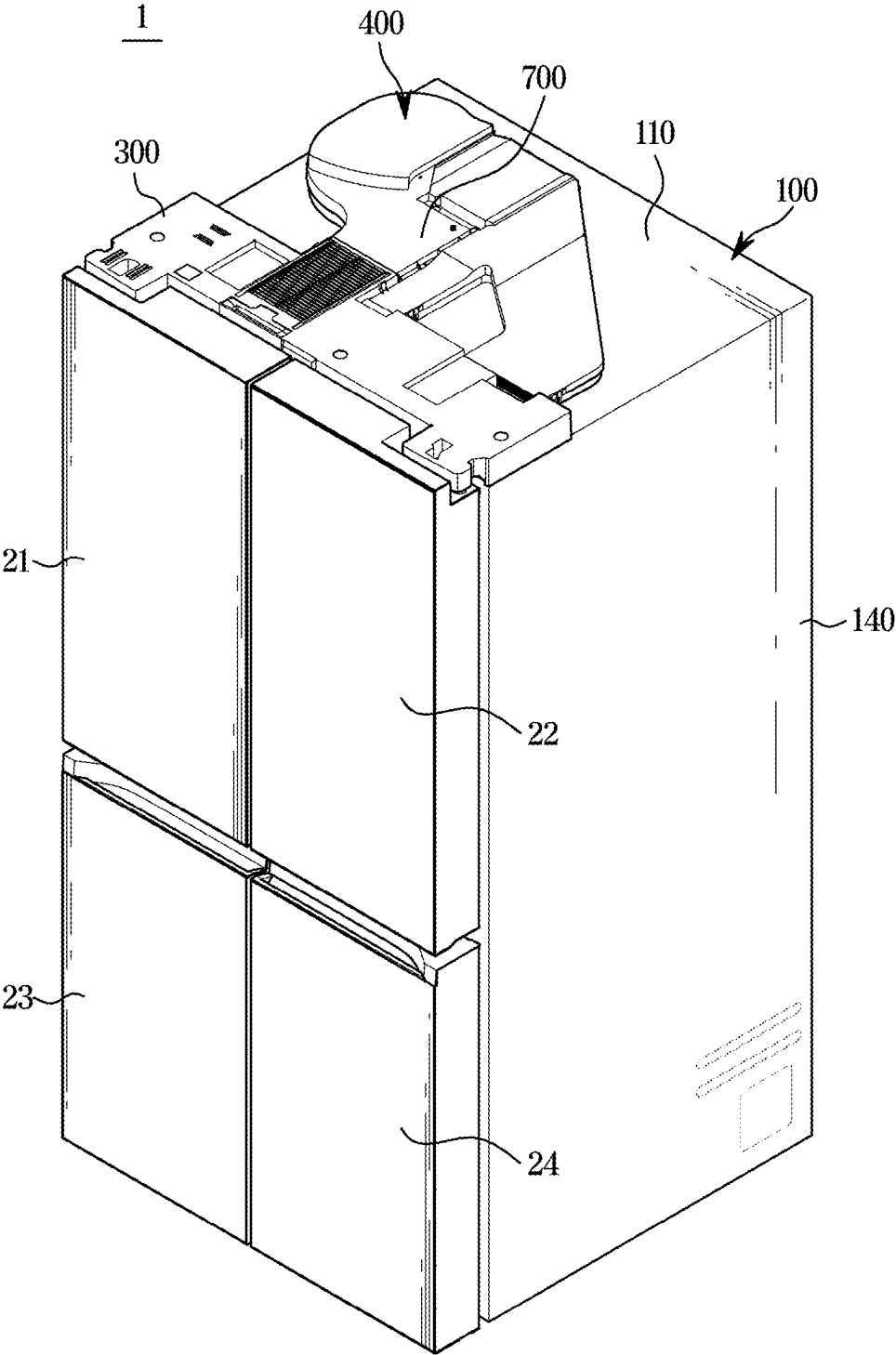


FIG. 3

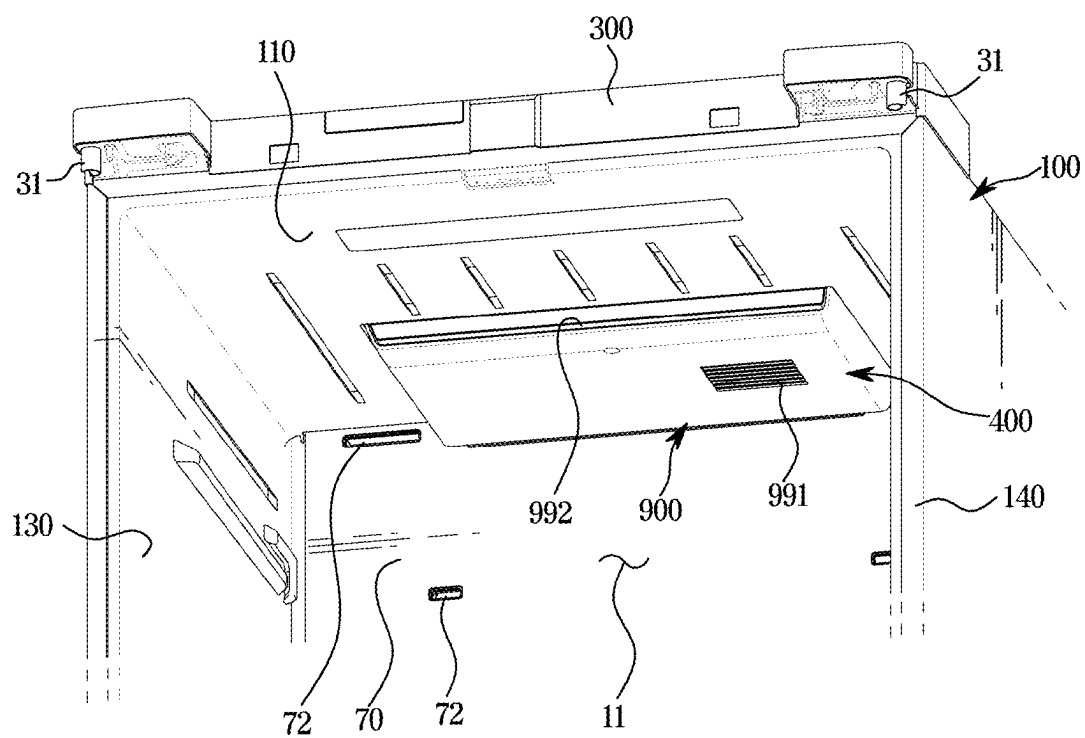


FIG. 4

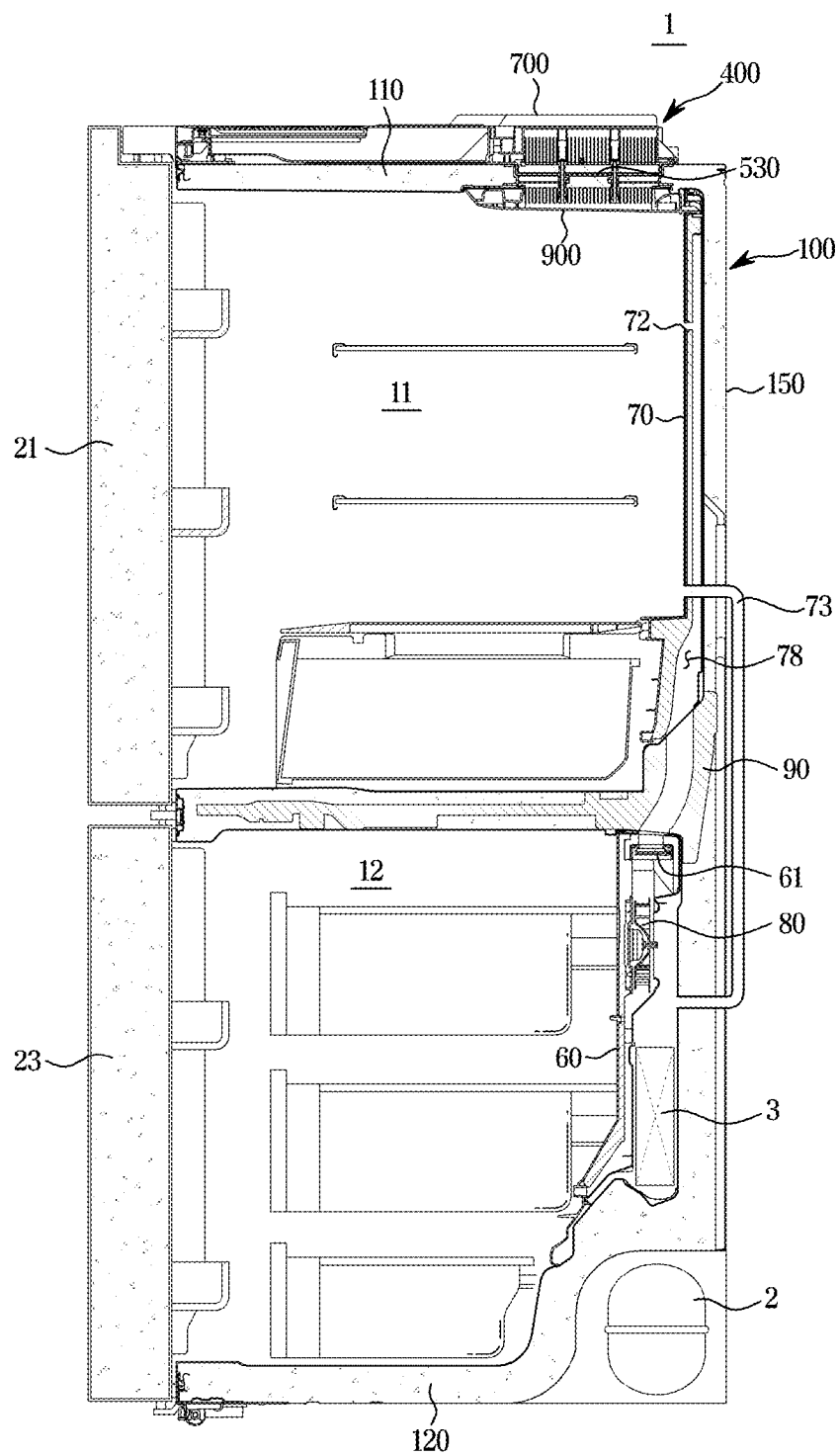


FIG. 5

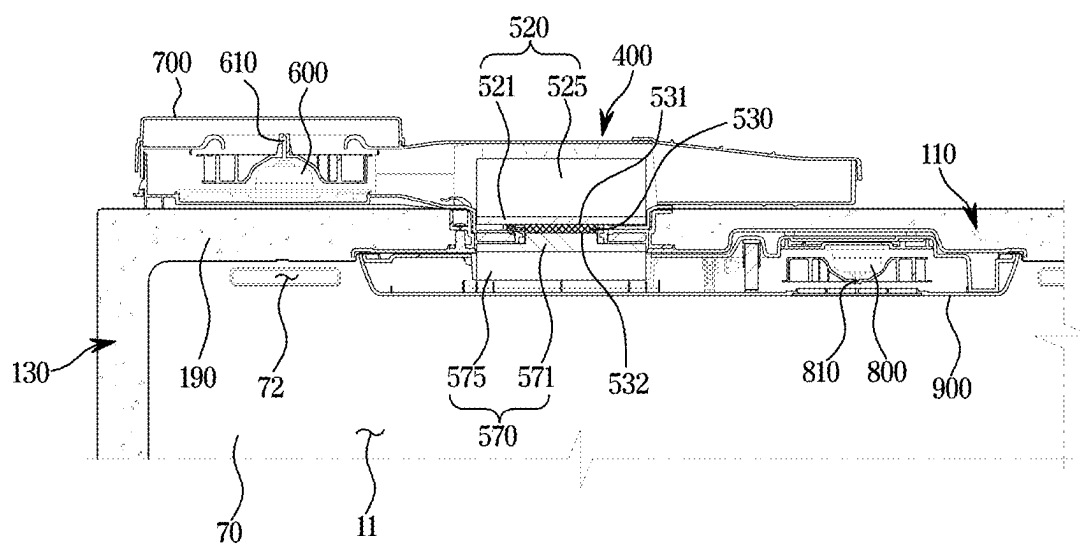


FIG. 6

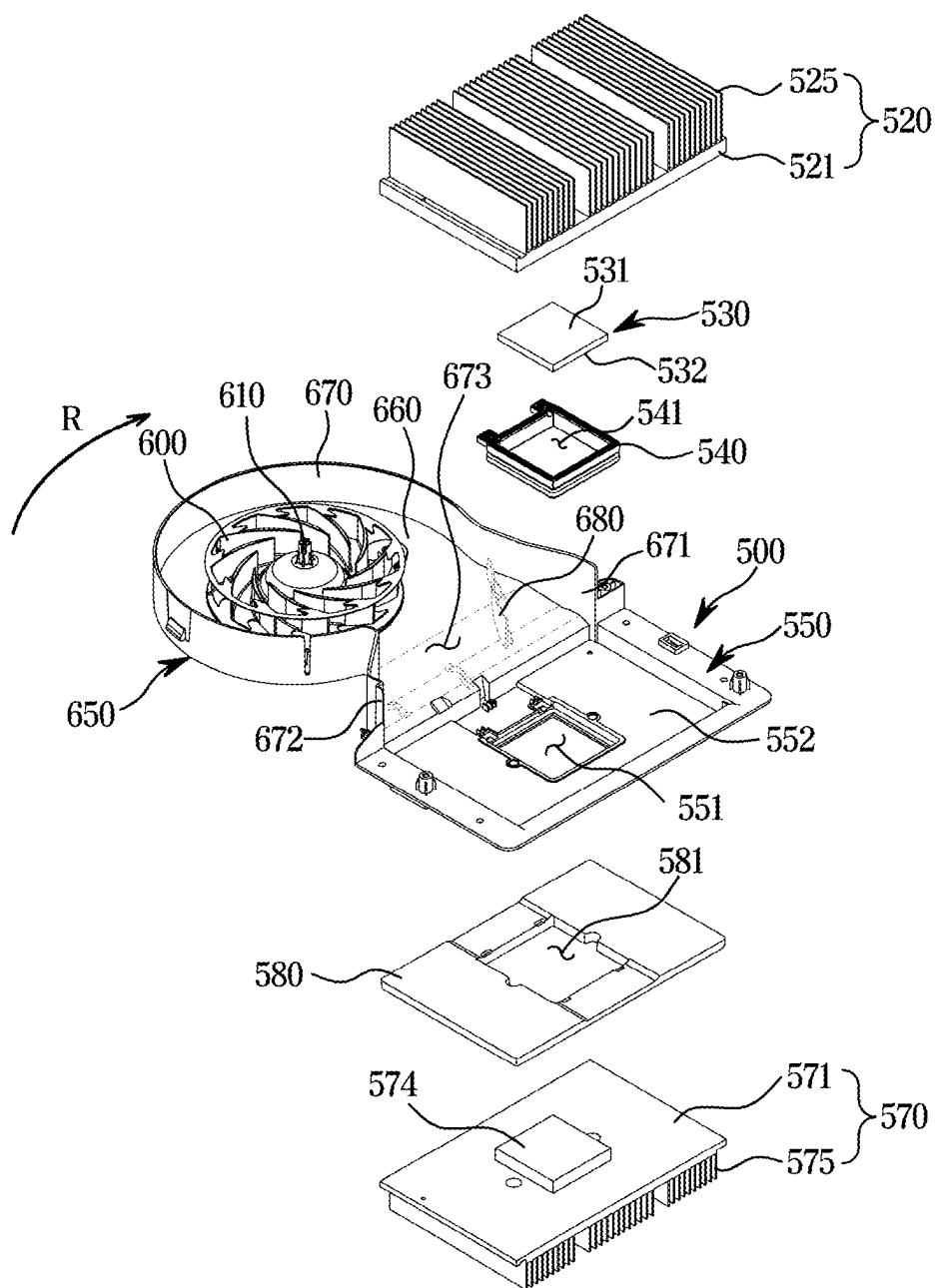


FIG. 7

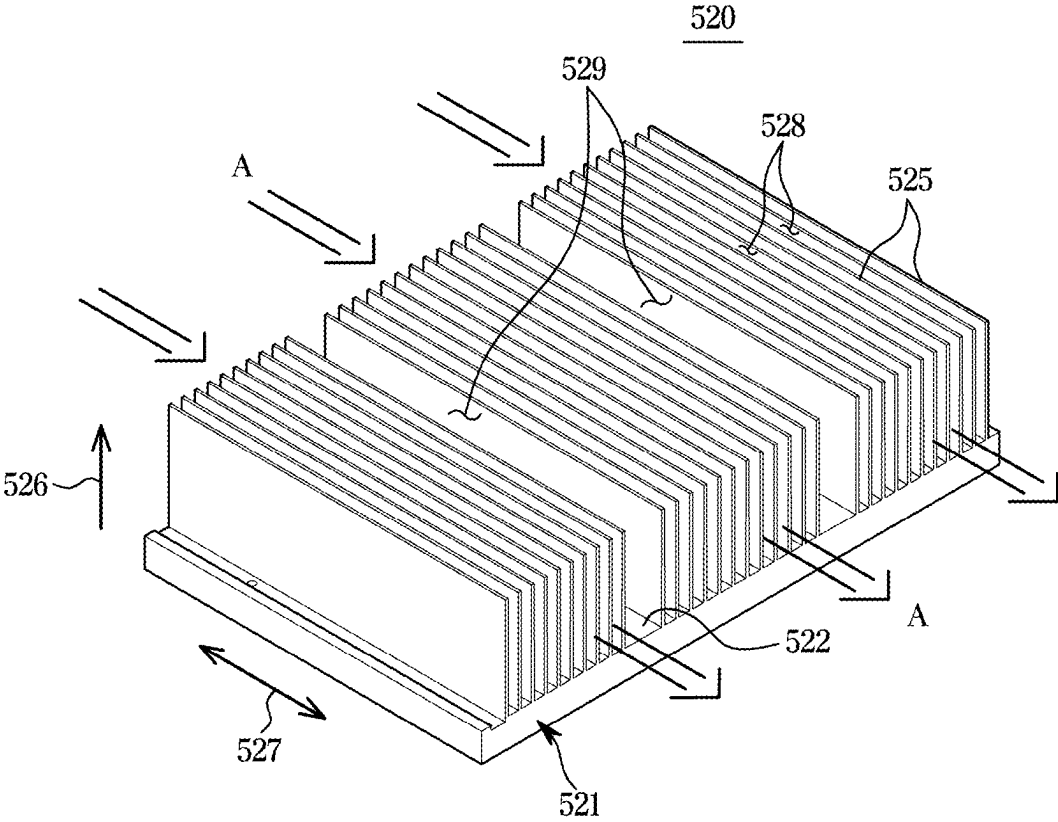


FIG. 8

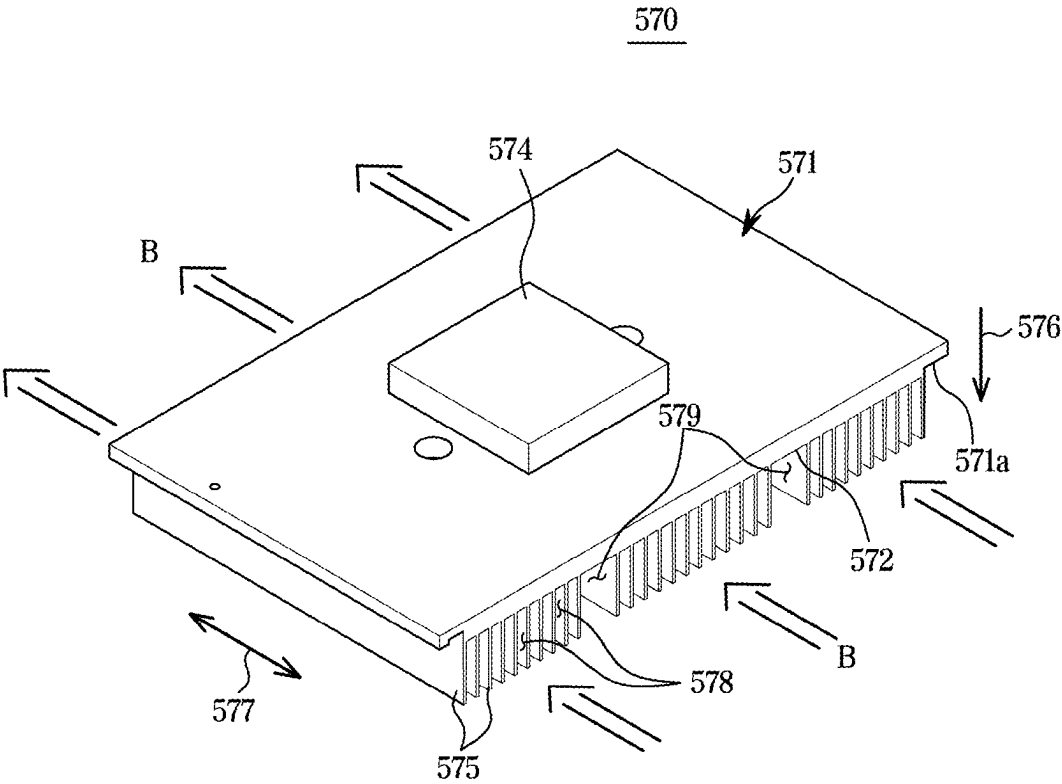


FIG. 9

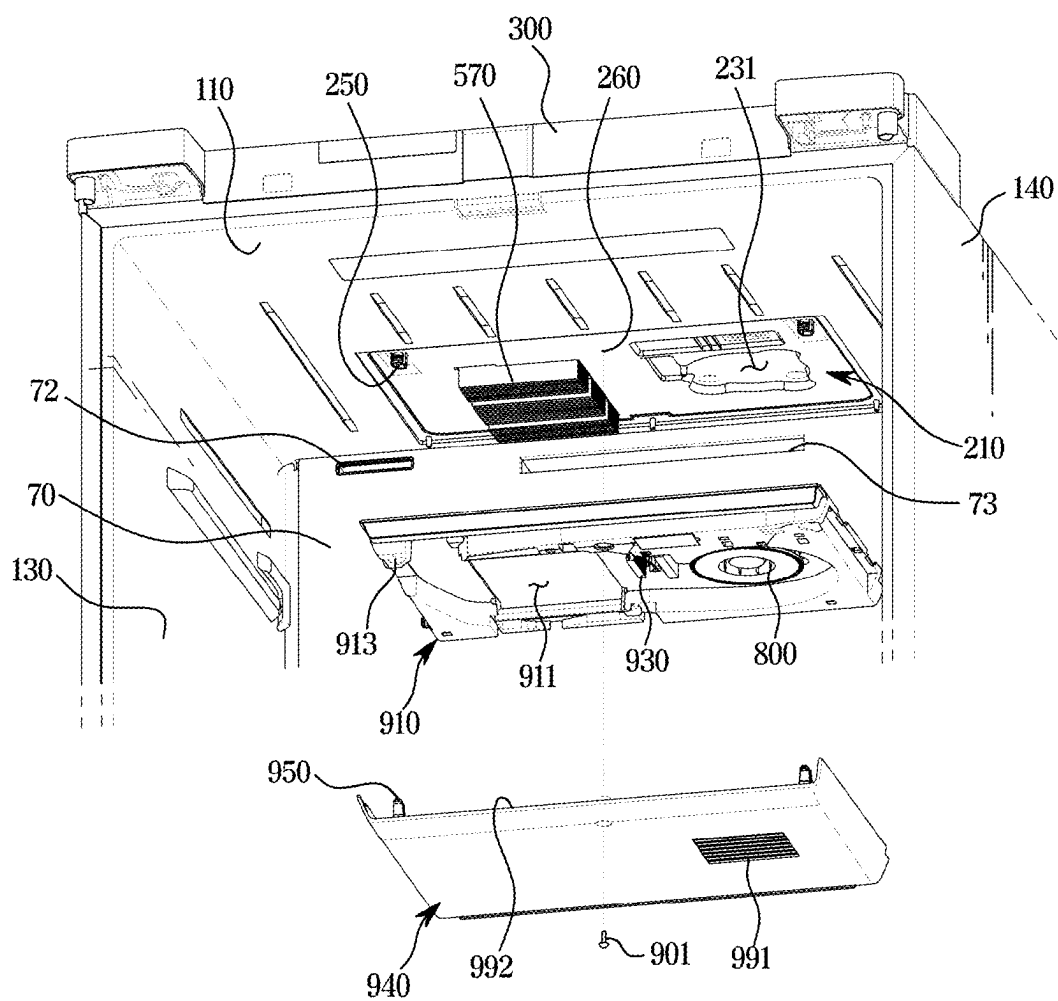


FIG. 10

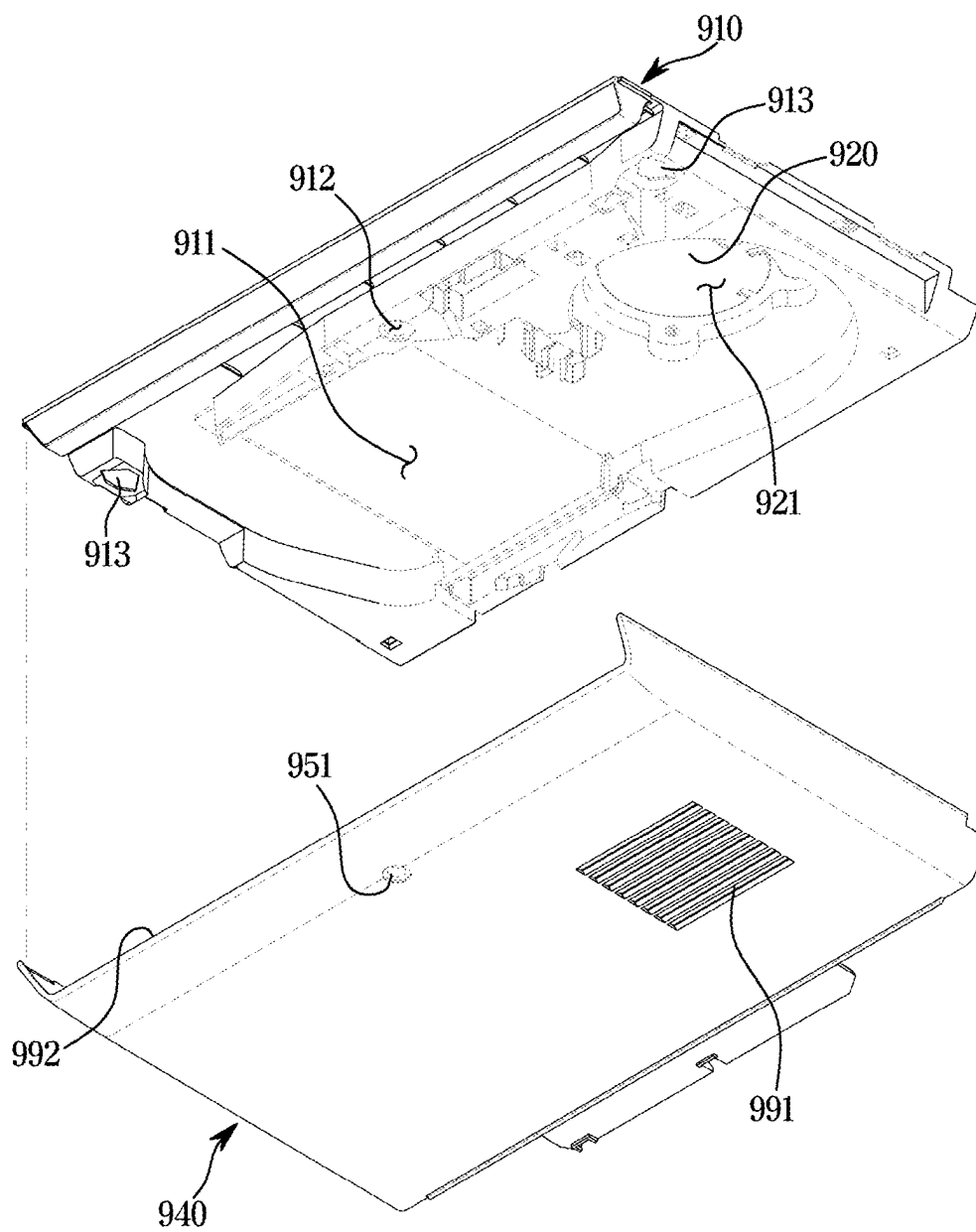


FIG. 11

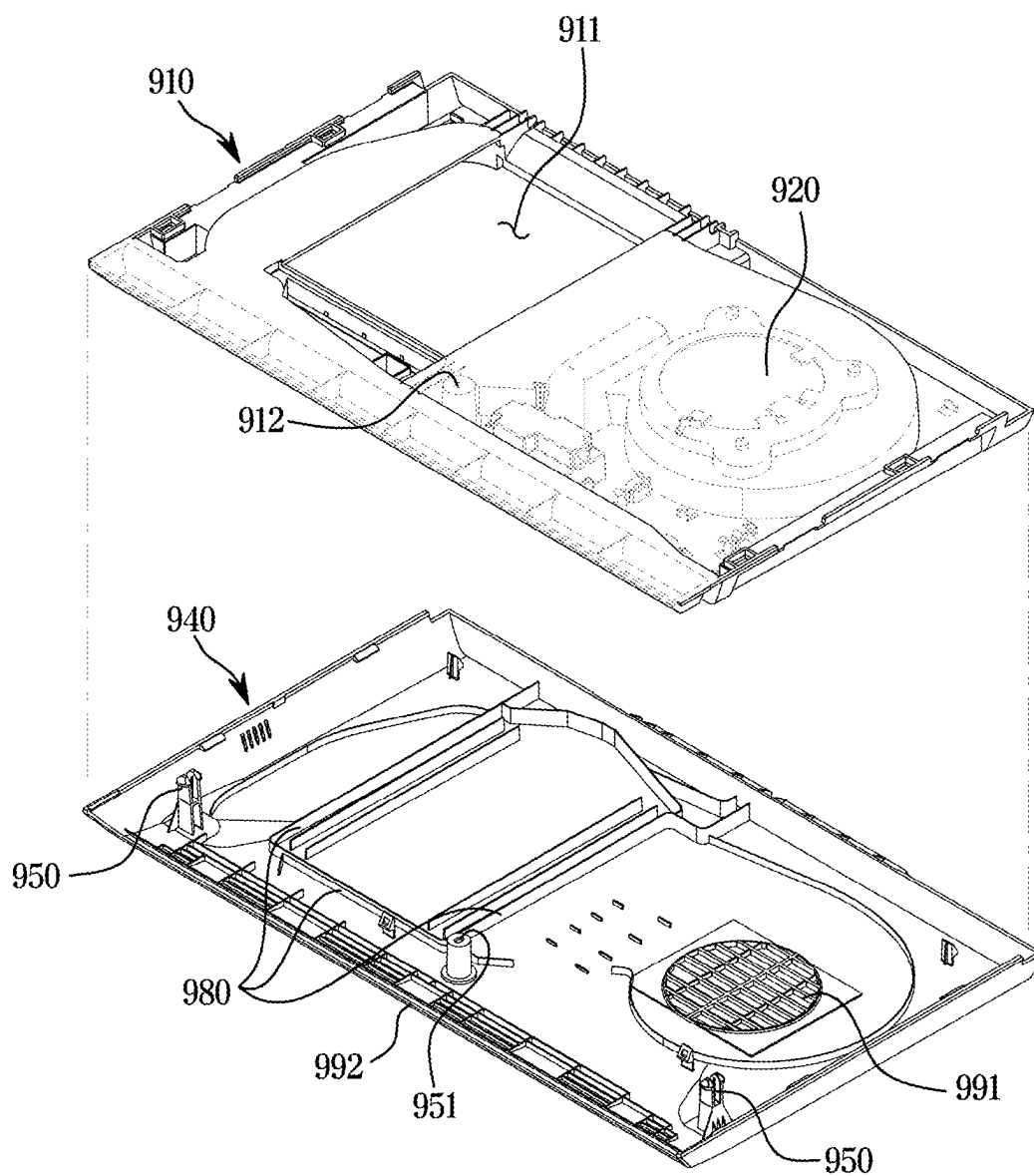


FIG. 12

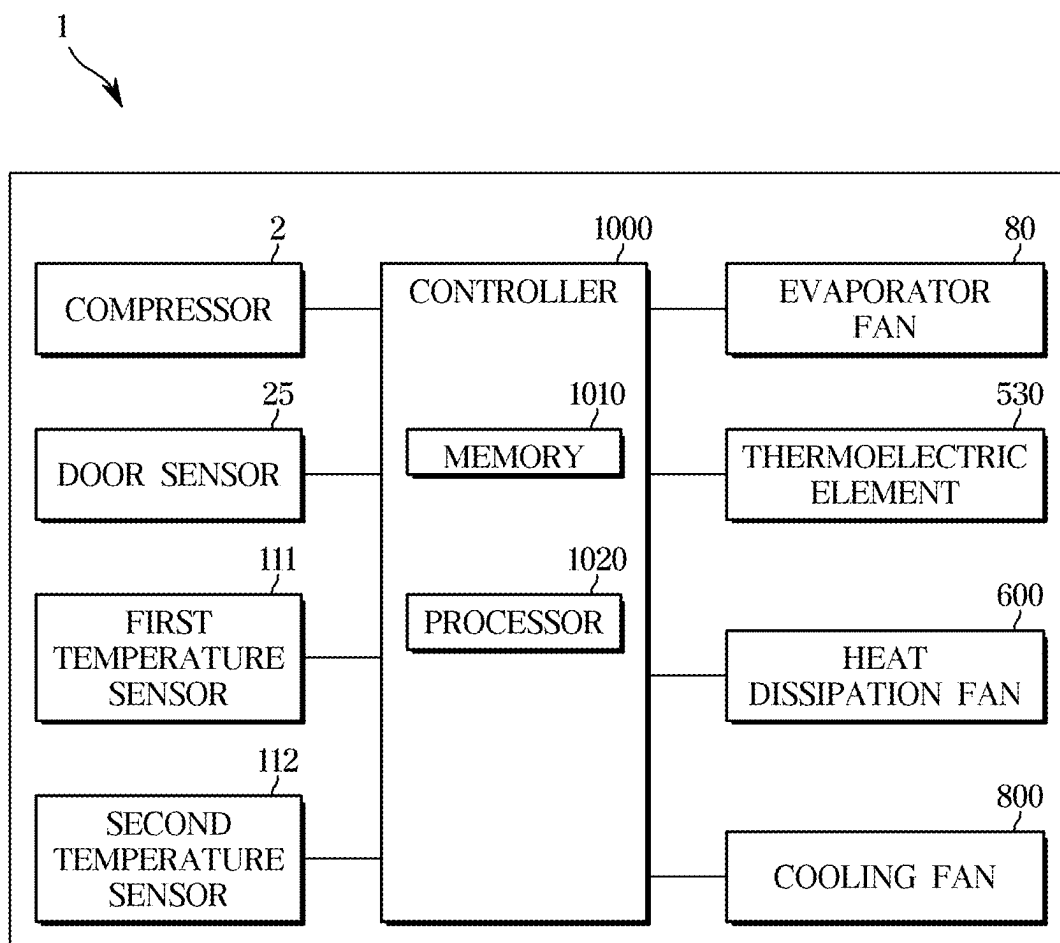


FIG. 13

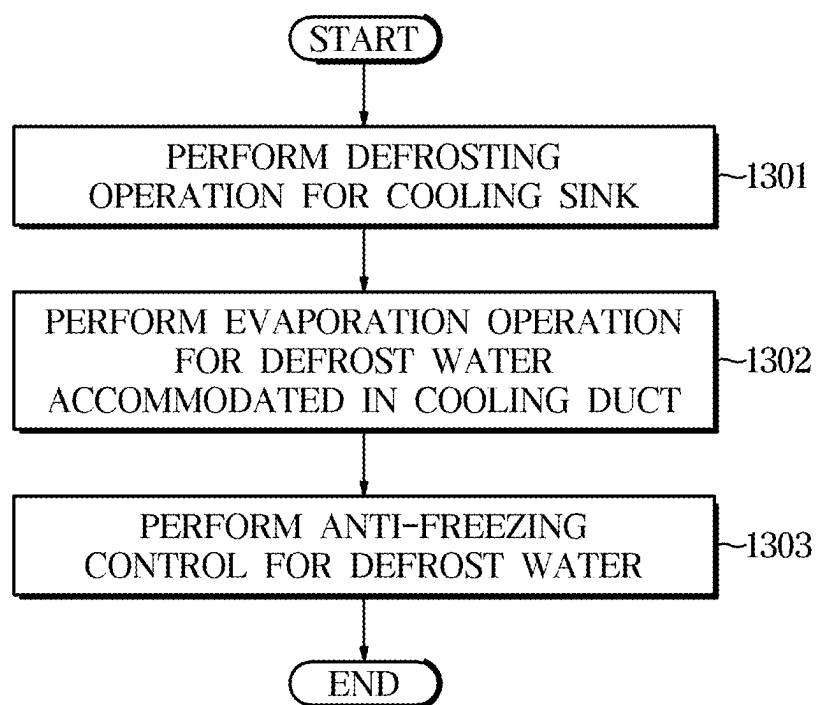


FIG. 14

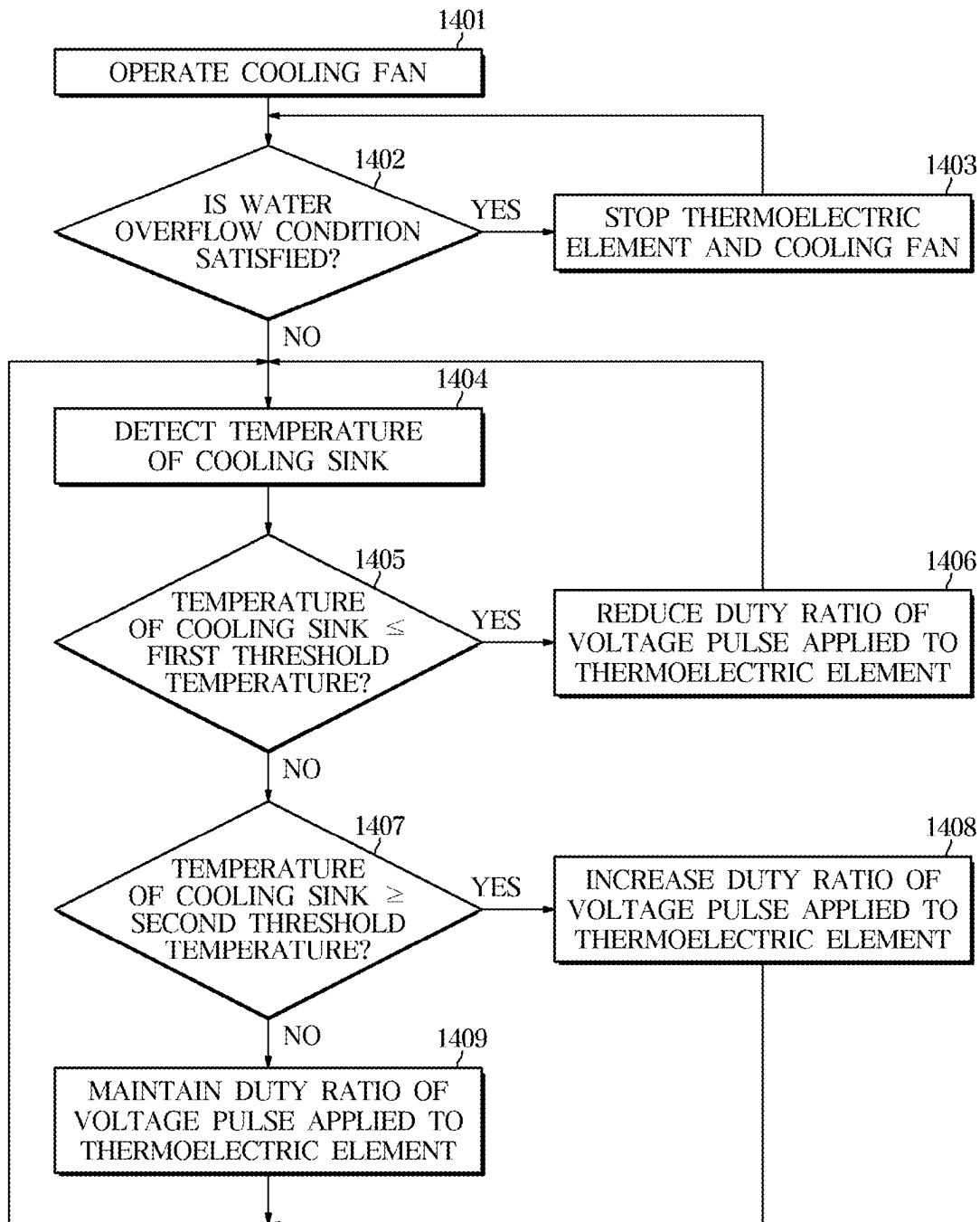


FIG. 15

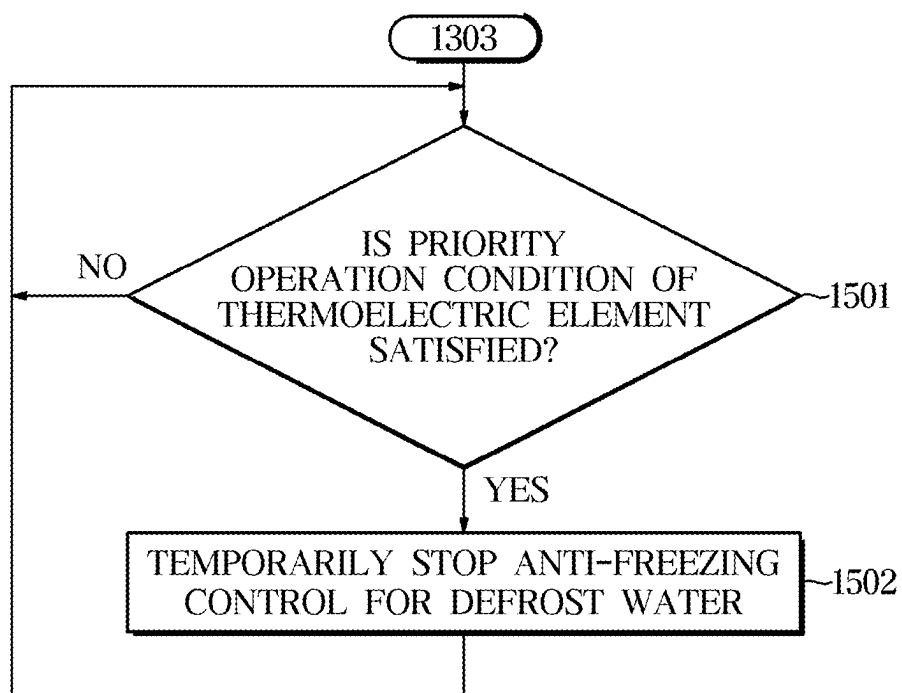


FIG. 16

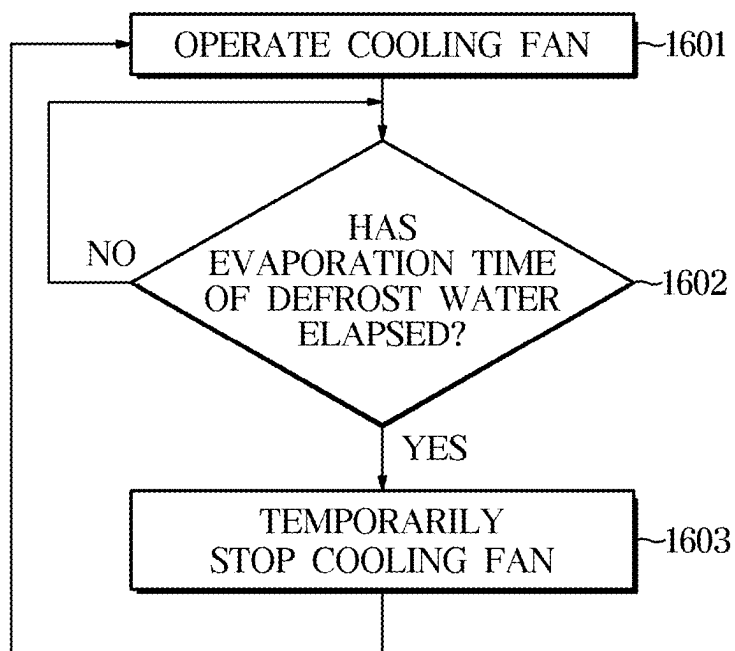
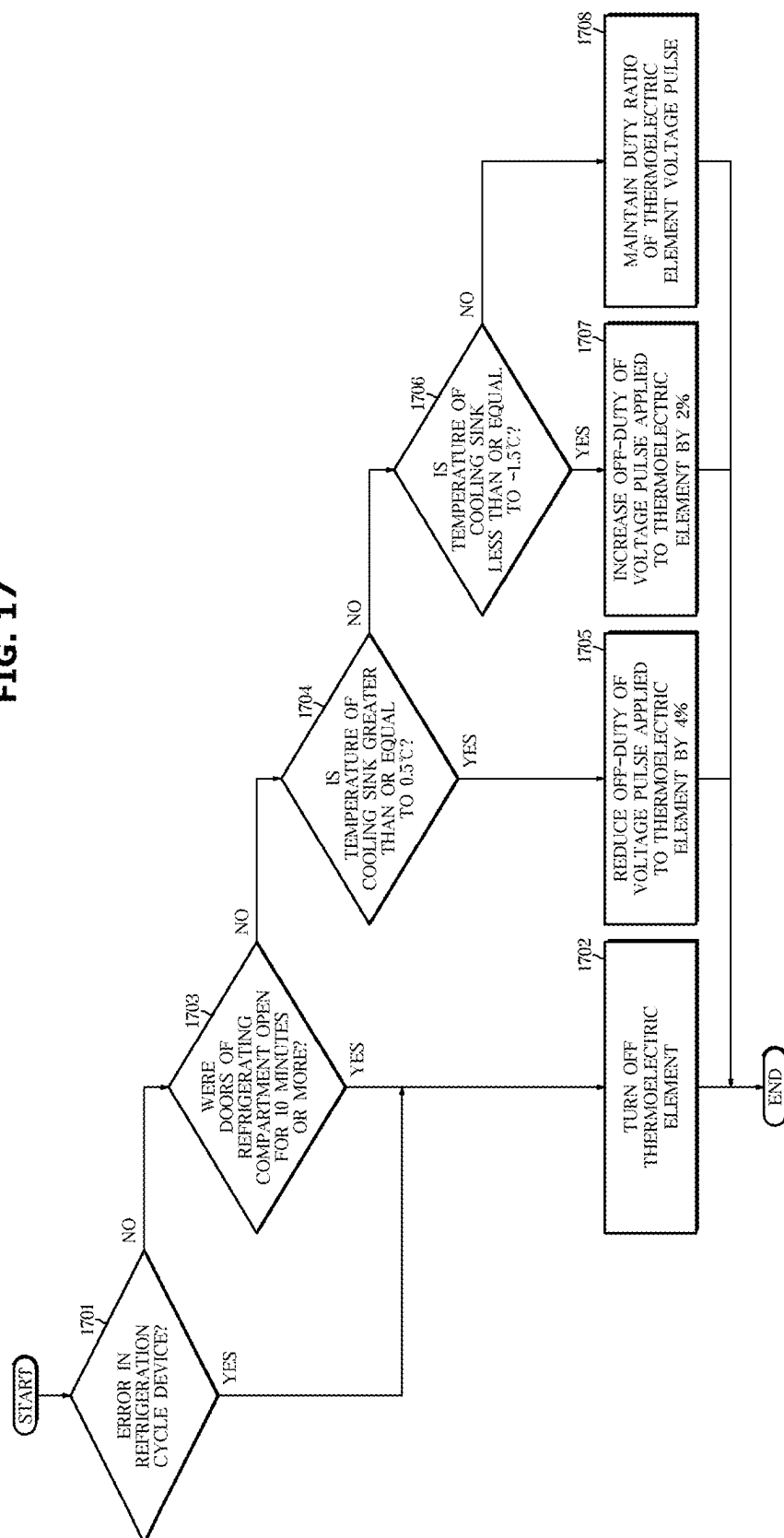


FIG. 17



REFRIGERATOR AND METHOD FOR CONTROLLING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This is a continuation application, under 35 U.S.C. § 111(a), of International Application PCT/KR2025/002599, filed Feb. 25, 2025, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0031628, filed Mar. 5, 2024 and Korean Patent Application No. 10-2024-0078561, filed Jun. 17, 2024, in the Korean Intellectual Property Office, the disclosures of which are incorporated herein in their entireties by reference.

TECHNICAL FIELD

[0002] The disclosure relates to a refrigerator, and more particularly, to a refrigerator including a thermoelectric element to cool a storage compartment.

BACKGROUND ART

[0003] A refrigerator is an appliance that is equipped with a main body having a storage compartment and a cold air supply device for supplying the storage compartment with cold air to store food in a fresh state.

[0004] A thermoelectric cooler that generates heating and cooling by the Peltier effect may be used as a cold air supply device of a refrigerator. The thermoelectric cooler may include a thermoelectric element. The thermoelectric element has a heating layer formed on one side and a cooling layer formed on the opposite side, and when current is applied to the thermoelectric element, heating may occur in the heating layer and heat absorption may occur in the cooling layer.

[0005] The thermoelectric cooler may include a heat sink, a cooling sink, a heat dissipation fan, a cooling fan, a heat dissipation duct, and a cooling duct to increase the cooling efficiency of a storage compartment through the thermoelectric cooler.

DISCLOSURE

[0006] The disclosure provides a refrigerator including a thermoelectric cooler using a thermoelectric element, and a method for controlling the same.

[0007] The disclosure provides a refrigerator that may defrost a cooling sink included in a thermoelectric cooler to remove defrost water contained in a cooling duct and prevent the defrost water from freezing, and a method for controlling the same.

[0008] The disclosure provides a refrigerator that may prevent water overflow in a cooling duct, and a method for controlling the same.

[0009] In accordance with the present disclosure, a refrigerator may include: a storage compartment; a cooling sink; a thermoelectric element including a cooling layer in contact with the cooling sink, the thermoelectric element configured to, based on a voltage pulse applied to the thermoelectric element, cool the cooling layer and thereby cool the cooling sink; a cooling fan controllable to generate a flow of air from the storage compartment to the cooling sink to be cooled by the cooling sink; a cooling duct covering the cooling sink and the cooling fan, and configured to guide the flow of air from the storage compartment to the cooling sink and to accommodate defrost water generated by water, frozen by

the cooling sink, being defrosted; a temperature sensor configured to detect a temperature of the cooling sink; and a processor configured to: control the cooling fan to evaporate the defrost water accommodated in the cooling duct, and perform an anti-freezing control so that the defrost water accommodated in the cooling duct does not freeze, the anti-freezing control including adjusting a duty ratio of the voltage pulse based on the temperature of the cooling sink detected by the temperature sensor.

[0010] The processor may be further configured to determine the duty ratio based on the temperature of the cooling sink detected by the temperature sensor over a time interval so that the temperature of the cooling sink detected by the temperature sensor is maintained within a temperature range which prevents the defrost water accommodated in the cooling duct from freezing.

[0011] The processor may be further configured to: reduce the duty ratio based on the temperature of the cooling sink detected by the temperature sensor being lower than or equal to a first threshold temperature; and increase the duty ratio based on the temperature of the cooling sink detected by the temperature sensor being greater than or equal to a second threshold temperature that is greater than the first threshold temperature.

[0012] The processor may be further configured to: maintain the duty ratio based on the temperature of the cooling sink detected by the temperature sensor being greater than a first threshold temperature and lower than a second threshold temperature, and the first threshold temperature may be lower than the second threshold temperature.

[0013] The processor may be further configured to, based on a door of the refrigerator being opened for a threshold time, not apply the voltage pulse to the thermoelectric element and control the cooling fan to not generate the flow of air so that the defrost water accommodated in the cooling duct does not overflow the cooling duct.

[0014] The refrigerator may further include: an evaporator outside the storage compartment and configured to cool air to be supplied to the storage compartment; and a compressor configured to supply a refrigerant to the evaporator, and the processor may be further configured to, based on refrigerant not being supplied to the evaporator by the compressor, not apply the voltage pulse to the thermoelectric element and control the cooling fan to not generate the flow of air.

[0015] The refrigerator may further include: an evaporator outside the storage compartment and configured to cool air to be supplied to the storage compartment; and a return duct configured to guide air to be cooled from the storage compartment to the evaporator.

[0016] The refrigerator may further include: an evaporator fan configured to move air cooled by the evaporator to the storage compartment, wherein the processor may be further configured to temporarily control the cooling fan to not generate the flow of air so that moisture in the storage compartment is allowed to move to the evaporator through the return duct by operation of the evaporator fan, based on elapse of an evaporation time for evaporating the defrost water after the cooling fan is controlled to generate the flow or air.

[0017] The processor may be further configured to temporarily stop the anti-freezing control based on a priority operation condition of the thermoelectric element being satisfied.

[0018] In accordance with the present disclosure, a method for controlling a refrigerator including a storage compartment, a cooling sink, a thermoelectric element including a cooling layer in contact with the cooling sink, the thermoelectric element configured to, based on a voltage pulse applied to the thermoelectric element, cool the cooling layer and thereby cool the cooling sink, the refrigerator further including a cooling fan controllable to generate a flow of air from the storage compartment to the cooling sink to be cooled by the cooling sink, a cooling duct covering the cooling sink and the cooling fan, and configured to guide the flow of air from the storage compartment to the cooling sink and to accommodate defrost water generated by water, frozen by the cooling sink, being defrosted, a temperature sensor configured to detect a temperature of the cooling sink, and a processor, the method may include: by the processor, controlling the cooling fan to evaporate the defrost water accommodated in the cooling duct; and performing an anti-freezing control so that the defrost water accommodated in the cooling duct does not freeze, the anti-freezing control including adjusting a duty ratio of a voltage pulse based on the temperature of the cooling sink detected by the temperature sensor.

[0019] The performing of the anti-freezing control may further include, by the processor, determining the duty ratio based on the temperature of the cooling sink detected by the temperature sensor over a time interval so that the temperature of the cooling sink detected by the temperature sensor is maintained within a temperature range which prevents the defrost water accommodated in the cooling duct from freezing.

[0020] The performing of the anti-freezing control may further include: reducing the duty ratio based on the temperature of the cooling sink detected by the temperature sensor being lower than or equal to a first threshold temperature, and increasing the duty ratio based on the temperature of the cooling sink detected by the temperature sensor being greater than or equal to a second threshold temperature that is greater than the first threshold temperature.

[0021] The performing of the anti-freezing control may further include: maintaining the duty ratio based on the temperature of the cooling sink detected by the temperature sensor being greater than a first threshold temperature and lower than a second threshold temperature, and the first threshold temperature is lower than the second threshold temperature.

[0022] The method may further include: by the processor, based on a door of the refrigerator being opened for a threshold time, not applying the voltage pulse to the thermoelectric element and controlling the cooling fan to not generate the flow of air so that the defrost water accommodated in the cooling duct does not overflow the cooling duct.

[0023] The refrigerator may further include: an evaporator outside the storage compartment and configured to cool air to be supplied to the storage compartment, and a compressor configured to supply a refrigerant to the evaporator, and the method may further include, by the processor, based on refrigerant not being supplied to the evaporator by the compressor, not applying the voltage pulse to the thermoelectric element, and controlling the cooling fan to not generate the flow of air.

[0024] According to the disclosure, a refrigerator and a method for controlling the same may increase a cooling efficiency of a storage compartment using a thermoelectric cooler.

[0025] According to the disclosure, the refrigerator and the method for controlling the same may remove defrost water generated by defrosting a cooling sink of a thermoelectric cooler, thereby preventing a performance of the thermoelectric cooler from deteriorating.

[0026] According to the disclosure, the refrigerator and the method for controlling the same may prevent defrost water accommodated in a cooling duct from freezing.

[0027] According to the disclosure, the refrigerator and the method for controlling the same may prevent water overflow in a cooling duct.

[0028] The effects that may achieved by the disclosure are not limited to the above-mentioned effects, and other effects not mentioned will be clearly understood by one of ordinary skill in the technical art to which the disclosure belongs from the following description.

DESCRIPTION OF DRAWINGS

[0029] FIG. 1 illustrates a refrigerator according to an embodiment.

[0030] FIG. 2 is a view illustrating a state in which doors of a refrigerator are opened according to an embodiment.

[0031] FIG. 3 is a view illustrating an upper part of a storage compartment of a refrigerator, viewed from below, according to an embodiment.

[0032] FIG. 4 is a schematic side cross-sectional view of a refrigerator according to an embodiment.

[0033] FIG. 5 is a cross-sectional view along I-I' of FIG. 2.

[0034] FIG. 6 is an exploded view of a heat dissipation fan and a thermoelectric module according to an embodiment.

[0035] FIG. 7 illustrates a heat sink according to an embodiment.

[0036] FIG. 8 illustrates a cooling sink according to an embodiment.

[0037] FIG. 9 is a view illustrating a state in which a cooling duct cover and a cooling duct body are separated from an upper wall of a main body according to an embodiment.

[0038] FIG. 10 is an exploded view of a cooling duct cover and a cooling duct body, viewed from below, according to an embodiment.

[0039] FIG. 11 is an exploded view of a cooling duct cover and a cooling duct body, viewed from above, according to an embodiment.

[0040] FIG. 12 is a control block diagram of a refrigerator according to an embodiment.

[0041] FIG. 13 is a flowchart briefly illustrating a method for controlling a refrigerator according to an embodiment.

[0042] FIG. 14 is a flowchart illustrating a method for controlling a refrigerator according to an embodiment in greater detail.

[0043] FIG. 15 is a flowchart illustrating a method of temporarily stopping the anti-freezing control for defrost water described in FIG. 13.

[0044] FIG. 16 is a flowchart further illustrating the control of the cooling fan in the evaporation operation for defrost water described in FIG. 13.

[0045] FIG. 17 is a flowchart illustrating a detailed embodiment of the method for controlling the refrigerator described in FIG. 14.

MODES OF THE DISCLOSURE

[0046] Various embodiments of the disclosure and terms used herein are not intended to limit the technical features described herein to specific embodiments, and should be understood to include various modifications, equivalents, or substitutions of the corresponding embodiments.

[0047] In describing of the drawings, similar reference numerals may be used for similar or related elements.

[0048] The singular form of a noun corresponding to an item may include one or more of the items unless clearly indicated otherwise in a related context.

[0049] In the disclosure, phrases, such as “A or B”, “at least one of A and B”, “at least one of A or B”, “A, B or C”, “at least one of A, B and C”, and “at least one of A, B, or C” may include any one or all possible combinations of the items listed together in the corresponding phrase among the phrases.

[0050] As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0051] Terms such as “1st”, “2nd”, “primary”, or “secondary” may be used simply to distinguish an element from other elements, without limiting the element in other aspects (e.g., importance or order).

[0052] It will be understood that when the terms “includes”, “comprises”, “including”, and/or “comprising” are used in the disclosure, they specify the presence of the specified features, figures, steps, operations, components, members, or combinations thereof, but do not preclude the presence or addition of one or more other features, figures, steps, operations, components, members, or combinations thereof.

[0053] When a given element is referred to as being “connected to”, “coupled to”, “supported by” or “in contact with” another element, it is to be understood that it may be directly or indirectly connected to, coupled to, supported by, or in contact with the other element. When a given element is indirectly connected to, coupled to, supported by, or in contact with another element, it is to be understood that it may be connected to, coupled to, supported by, or in contact with the other element through a third element.

[0054] It will also be understood that when an element is referred to as being “on” another element, it may be directly on the other element or intervening elements may also be present.

[0055] The terms “front,” “rear,” “top,” “bottom,” “side,” “left,” “right,” “upper,” “lower”, and the like are defined with reference to the drawings, and are not intended to limit the shape and position of any element. For example, the front side may be defined as the +X side and the rear side may be defined as the -X side. For example, based on the drawings, the right side may be defined as the +Y side and the left side may be defined as the -Y side. For example, based on the drawings, the upper side may be defined as the +Z side and the lower side may be defined as the -Z side.

[0056] Further, the terms such as “~portion”, “~device”, “~block”, “~member”, “~module”, and the like may refer to a unit for processing at least one function or act. For example, the terms may refer to at least one process processed by at least one hardware or software stored in

memories or processors, such as field-programmable gate array (FPGA)/application specific integrated circuit (ASIC), software stored in memories or processors.

[0057] Hereinafter, an embodiment according to the disclosure will be described in detail with reference to the accompanying drawings.

[0058] A refrigerator according to an embodiment of the disclosure may include a main body.

[0059] The “main body” may include an insulation. The “insulation” may insulate an inside of the storage compartment from an outside of the storage compartment to maintain inside temperature of the storage compartment at appropriate temperature without being influenced by an external environment of the storage compartment. According to an embodiment of the disclosure, the insulation may include a foaming insulation such as polyurethane foam. According to an embodiment, the insulation may include a vacuum insulation in addition to a foaming insulation, or may be configured only with a vacuum insulation instead of a forming insulation.

[0060] The “storage compartment” may store a variety of items, such as food, medicines, cosmetics, and the like, and the storage compartment may be configured to be open on at least one side for insertion and removal of the items.

[0061] The refrigerator may include one or more storage compartments. In a case in which two or more storage compartments are formed in the refrigerator, the respective storage compartments may have different purposes of use, and may be maintained at different temperatures. To this end, the respective storage compartments may be partitioned by a partition wall including an insulation.

[0062] The storage compartment may be maintained within an appropriate temperature range according to a purpose of use, and may include a “refrigerating compartment”, a “freezing compartment”, and a “temperature conversion compartment” according to purposes of use and/or temperature ranges. The refrigerating compartment may be maintained at an appropriate temperature to keep food refrigerating, and the freezing compartment may be maintained at an appropriate temperature to keep food frozen. The “refrigerating” may be keeping food cold without freezing the food, and for example, the refrigerating compartment may be maintained within a range of 0 degrees Celsius to 7 degrees Celsius. The “freezing” may be freezing food or keeping food frozen, and for example, the freezing compartment may be maintained within a range of -20 degrees Celsius to -1 degrees Celsius. The temperature conversion compartment may be used as either a refrigerating compartment or a freezing compartment according to or regardless of a user’s selection.

[0063] The storage compartment may also be referred to by various terms, such as “vegetable compartment”, “freshness compartment”, “cooling compartment”, and “ice-making compartment”, in addition to “refrigerating compartment”, “freezing compartment”, and “temperature conversion compartment”, and the terms, such as “refrigerating compartment”, “freezing compartment”, “temperature conversion compartment”, etc., as used below are to be understood as representing storage compartments having the corresponding purposes of use and the corresponding temperature ranges.

[0064] The refrigerator according to an embodiment of the disclosure may include at least one door configured to open or close the open side of the storage compartment. The

respective doors may be provided to open and close one or more storage compartments, or a single door may be provided to open and close a plurality of storage compartments. The door may be rotatably or slidably mounted to the front of the main body.

[0065] The “door” may seal the storage compartment in a closed state. The door, like the main body, may include an insulation to insulate the storage compartment in a closed state.

[0066] According to an embodiment, the door may include an outer door plate forming the front surface of the door, an inner door plate forming the rear surface of the door and facing the storage compartment, an upper cap, a lower cap, and a door insulation provided therein.

[0067] A gasket may be provided on the edge of the inner door plate to seal the storage compartment by coming into close contact with the front surface of the main body when the door is closed. The inner doorplate may include a dyke that protrudes rearward to allow a door basket for storing items to be fitted.

[0068] According to an embodiment, the door may include a door body and a front panel that is detachably coupled to the front of the door body and forming the front surface of the door. The door body may include an outer door plate forming the front surface of the door body, an inner door plate forming the rear surface of the door body and facing the storage compartment, an upper cap, a lower cap, and a door insulator provided therein.

[0069] The refrigerator may be classified as French Door Type, Side-by-side Type, Bottom Mounted Freezer (BMF), Top Mounted Freezer (TMF), or Single Door Refrigerator according to the arrangement of the doors and the storage compartments.

[0070] The refrigerator according to an embodiment of the disclosure may include a cold air supply device for supplying cold air to the storage compartment.

[0071] The “cold air supply device” may include a machine, an apparatus, an electronic device, and/or a combination system thereof, capable of generating cold air and guiding the cold air to cool the storage compartment.

[0072] According to an embodiment of the disclosure, the cold air supply device may generate cold air through a cooling cycle including compression, condensation, expansion, and evaporation processes of refrigerants. To this end, the cold air supply device may include a refrigeration cycle device having a compressor, a condenser, an expander, and an evaporator to drive the refrigeration cycle. According to an embodiment of the disclosure, the cold air supply device may include a semiconductor, such as a thermoelectric element. The thermoelectric element may cool the storage compartment by heating and cooling actions through the Peltier effect.

[0073] The refrigerator according to an embodiment of the disclosure may include a machine compartment in which at least some components belonging to the cold air supply device are installed.

[0074] The “machine compartment” may be partitioned and insulated from the storage compartment to prevent heat generated by the components installed in the machine compartment from being transferred to the storage compartment. To dissipate heat from the components installed in the machine compartment, the machine compartment may communicate with outside of the main body.

[0075] The refrigerator according to an embodiment of the disclosure may include a dispenser provided on the door to provide water and/or ice. The dispenser may be provided on the door to allow access by the user without opening the door.

[0076] The refrigerator according to an embodiment of the disclosure may include an ice-making device that produces ice. The ice-making device may include an ice-making tray that stores water, an ice-moving device that separates ice from the ice-making tray, and an ice-bucket that stores ice produced in the ice-making tray.

[0077] The refrigerator according to an embodiment of the disclosure may include a controller for controlling the refrigerator.

[0078] The “controller” may include a memory for storing and/or recording data and/or programs for controlling the refrigerator, and a processor for outputting control signals for controlling the cold air supply device, etc., in accordance with the programs and/or data stored in the memory.

[0079] The memory may store or record various information, data, instructions, programs, and the like necessary for operation of the refrigerator. The memory may store temporary data generated while generating control signals for controlling components included in the refrigerator.

[0080] The memory may include at least one of a volatile memory or a non-volatile memory, or a combination thereof.

[0081] The processor may control the overall operation of the refrigerator. The processor may control the components of the refrigerator by executing programs stored in memory. The processor may include a separate neural processing unit (NPU) that performs an artificial intelligence (AI) model operation. In addition, the processor may include a central processing unit (CPU), a graphics processor (GPU), and the like. The processor may generate a control signal to control the operation of the cold air supply device. For example, the processor may receive temperature information of the storage compartment from a temperature sensor and generate a cooling control signal to control an operation of the cold air supply device based on the temperature information of the storage compartment.

[0082] Furthermore, the processor may process a user input of a user interface and control an operation of the user interface in accordance with the programs and/or data memorized/stored in the memory. The user interface may be provided with an input interface and an output interface. The processor may receive the user input from the user interface. In addition, the processor may transmit a display control signal and image data for displaying an image on the user interface to the user interface in response to the user input.

[0083] The processor and memory may be provided integrally or may be provided separately. The processor may include one or more processors. For example, the processor may include a main processor and at least one sub-processor. The memory may include one or more memories.

[0084] The refrigerator according to an embodiment of the disclosure may include a processor and a memory for controlling all of the components included in the refrigerator, and may include a plurality of processors and a plurality of memories for individually controlling the components of the refrigerator. For example, the refrigerator may include a processor and a memory for controlling the operation of the cold air supply device in accordance with to an output of the temperature sensor. In addition, the refrigerator may be

separately provided with a processor and a memory for controlling the operation of the user interface in accordance with the user input.

[0085] A communication module may communicate with external devices, such as servers, mobile devices, and other home appliances via a nearby access point (AP). The AP may connect a local area network (LAN) to which a refrigerator or a user device is connected to a wide area network (WAN) to which a server is connected. The refrigerator or the user device may be connected to the server via the WAN.

[0086] The input interface may include keys, a touch screen, a microphone, and the like. The input interface may receive the user input and pass the received user input to the processor.

[0087] The output interface may include a display, a speaker, and the like. The output interface may output various notifications, messages, information, and the like generated by the processor.

[0088] Hereinafter, an operation principle and embodiments according to the disclosure will be described in detail with reference to the accompanying drawings.

[0089] FIG. 1 illustrates a refrigerator according to an embodiment. FIG. 2 is a view illustrating a state in which doors of a refrigerator are opened according to an embodiment. FIG. 3 is a view illustrating an upper part of a storage compartment of a refrigerator, viewed from below, according to an embodiment. FIG. 4 is a schematic side cross-sectional view of a refrigerator according to an embodiment. FIG. 5 is a cross-sectional view along I-I' of FIG. 2.

[0090] Referring to FIG. 1 to FIG. 5, a refrigerator 1 may include a main body 100, storage compartments 11, 12, and 13 formed inside the main body 100, and doors 21, 22, 23, and 24 for opening and closing the storage compartments 11, 12, and 13. The main body 100 may include an upper wall 110, a lower wall 120, a left wall 130, a right wall 140, and a rear wall 150. The upper wall 110, the lower wall 120, the left wall 130, the right wall 140, and the rear wall 150 may form an upper side, a lower side, a left side, a right side, and a rear side of the main body 100, respectively.

[0091] Each of the upper wall 110, lower wall 120, left wall 130, right wall 140, and rear wall 150 may include an insulation 190. For example, the insulation 190 may be provided inside the upper wall 110. The storage compartments 11, 12, and 13 may be formed by the upper wall 110, the lower wall 120, the left wall 130, the right wall 140, and the rear wall 150.

[0092] The storage compartments 11, 12, and 13 may accommodate items. The storage compartments 11, 12, and 13 may be formed so that the front is open to allow items to be put in or taken out. The main body 100 may include a horizontal partition wall 160 separating the first storage compartment 11 from the second storage compartment 12 and the third storage compartment 13, and a vertical partition wall 161 separating the second storage compartment 12 from the third storage compartment 13. The first storage compartment 11 may be formed in an upper part of the main body 100, and the second storage compartment 12 and the third storage compartment 13 may be formed in a lower part of the main body 100. The first storage compartment 11 may be a refrigerating compartment. The second storage compartment 12 may be a freezing compartment. The third storage compartment 13 may be a temperature conversion compartment.

[0093] The doors 21, 22, 23, and 24 may open and close the storage compartments 11, 12, and 13. The first door 21 and the second door 22 may open and close the first storage compartment 11, the third door 23 may open and close the second storage compartment 12, and the fourth door 24 may open and close the third storage compartment 13. The doors 21, 22, 23, and 24 may be rotatably coupled to the main body 100.

[0094] The doors 21, 22, 23, and 24 may be rotatably coupled to the main body 100 by hinges. For example, the first door 21 and the second door 22 may be rotatably coupled to the main body 100 by a hinge 31 disposed on the upper part of the main body 100 and a hinge disposed in the middle of the main body 100, respectively. The hinge 31 may include a hinge pin that protrudes vertically to form a rotation axis of the door. The hinge 31 may be covered by a top cover 300 provided to cover an upper front part of the main body 100.

[0095] A rotation bar 40 may be arranged on one of the first door 21 and the second door 22 to cover a gap between the first door 21 and the second door 22 when the first door 21 and the second door 22 are closed. The rotation bar 40 may be rotatably disposed on one of the first door 21 and the second door 22. The rotation bar 40 may have a rod shape elongated in a vertical direction. The rotation bar 40 may also be referred to as a pillar, a mullion, and the like.

[0096] A guide protrusion 46 may be disposed at an upper end of the rotation bar 40, and a rotating guide 119 that guides the rotation of the guide protrusion 46 may be disposed at the upper part of the main body 100.

[0097] The doors 21, 22, 23, and 24 may include a gasket 51. The gasket 51 may make close contact with a front side of the main body 100 when the doors 21, 22, 23, and 24 are closed. The doors 21, 22, 23, and 24 may each include a dyke 52 that protrudes rearward. A door shelf 53 capable of storing items may be mounted on the dyke 52. The rotation bar 40 may be rotatably installed on the dyke 52.

[0098] Although the number and arrangement of the storage compartments and the number and arrangement of the doors have been described above, the number and arrangement of the storage compartments and the number and arrangement of the doors of the refrigerator according to an embodiment of the disclosure are not limited.

[0099] The refrigerator 1 may include a refrigeration cycle device to cool the storage compartments through a cooling cycle. The refrigeration cycle device may include a compressor 2 that compresses refrigerant, a condenser that condenses the refrigerant, an expansion valve that expands the refrigerant condensed by the condenser, and an evaporator 3 that is installed at the rear of the storage compartment and cools surrounding air. The condenser and the evaporator 3 may be referred to as a heat exchanger. The evaporator 3 may be disposed outside the storage compartments 11, 12, and 13. For example, the evaporator 3 may be installed at the rear of the storage compartments 12 and 13.

[0100] The refrigerator 1 may include a first temperature sensor 111 for detecting a temperature of the evaporator 3. The first temperature sensor 111 may measure the temperature of the evaporator 3. The temperature of the evaporator 3 may indicate a temperature of air around the evaporator 3 and/or a surface temperature of the evaporator 3. The first temperature sensor 111 may be provided in the evaporator 3. The first temperature sensor 111 may be provided in the evaporator ducts 60 and 70.

[0101] The evaporator 3 may generate cold air supplied to the storage compartments 11, 12, and 13. The compressor 2 may supply refrigerant to the evaporator 3. The refrigerant flowing into the evaporator 3 may exchange heat with the air around the evaporator 3. Because the refrigerant flowing into the evaporator 3 absorbs heat of the surrounding air, the surrounding air that has exchanged heat with the refrigerant may be cooled. The cooled air may be supplied to the storage compartments 11, 12, and 13 according to an operation of an evaporator fan 80.

[0102] The refrigerator 1 may include the evaporator ducts 60 and 70 that guide the cold air generated in the evaporator 3 to the storage compartments 11, 12, and 13. The first evaporator duct 60 may be disposed at the rear of the second storage compartment 12 and the third storage compartment 13. The second evaporator duct 70 may be disposed at the rear of the first storage compartment 11.

[0103] As the evaporator fan 80 operates, cold air generated in the evaporator 3 may be drawn into the first evaporator duct 60. The cold air drawn into the first evaporator duct 60 may be discharged to the second storage compartment 12 or the third storage compartment 13 through a cold air outlet (not shown) formed at the front.

[0104] In addition, a damper 61 for opening or closing the first evaporator duct 60 may be disposed in the first evaporator duct 60. When the damper 61 is opened, the first evaporator duct 60 is opened, and cold air in the first evaporator duct 60 may be supplied to the second evaporator duct 70. The cold air drawn into the first evaporator duct 60 may be guided to an internal flow path 78 of the second evaporator duct 70.

[0105] A connection duct 90 may be arranged between the first evaporator duct 60 and the second evaporator duct 70 to connect the first evaporator duct 60 and the second evaporator duct 70.

[0106] The cold air introduced into the internal flow path 78 of the second evaporator duct 70 may be supplied to the first storage compartment 11 through a cold air outlet 72 formed on the front side of the second evaporator duct 70.

[0107] Alternatively, the first evaporator duct 60 and the second evaporator duct 70 may not be connected to each other. In this case, cold air generated in the evaporator 3 may be supplied directly to the second evaporator duct 70 without passing through the first evaporator duct 60.

[0108] Although it has been described that a single evaporator 3 is provided, the disclosure is not limited thereto. For example, a separate evaporator 3 for supplying cold air to the second evaporator duct 70 may be provided at the rear of the first storage compartment 11.

[0109] Referring to FIG. 4, the first storage compartment 11 and the evaporator 3 may be connected through a return duct 73. For example, the return duct 73 may penetrate an upper part of the rear wall 150 and a lower part of the rear wall 150. The upper part of the rear wall 150 may form the rear side of the first storage compartment 11. The lower part of the rear wall 150 may form a space in which the evaporator 3 is located. The air in the first storage compartment 11 may move to the evaporator 3 through the return duct 73. The moisture in the first storage compartment 11 may move to the evaporator 3 through the return duct 73 and may be condensed in the evaporator 3. One or more return ducts 73 may be provided.

[0110] The refrigerator 1 may include a thermoelectric cooler 400 that may cool the first storage compartment 11.

The thermoelectric cooler 400 may be disposed above the first storage compartment 11. The thermoelectric cooler 400 may be disposed on the upper wall 110 of the main body 100.

[0111] The thermoelectric cooler 400 may include a thermoelectric element 530. The thermoelectric element 530 may convert thermal energy into electrical energy using the thermoelectric effect. In addition, the thermoelectric element 530 may convert electrical energy into thermal energy. The thermoelectric element 530 may be a semiconductor device. The thermoelectric element 530 may be referred to by various terms, such as a semiconductor thermoelectric element and a Peltier element. The thermoelectric element 530 may be disposed on the upper wall 110.

[0112] The thermoelectric element 530 includes a heating layer 531 and a cooling layer 532. When voltage and/or current is applied to the thermoelectric element 530, heating may occur in the heating layer 531 and heat absorption may occur in the cooling layer 532. The thermoelectric element 530 may have a thin hexahedral shape. The heating layer 531 may be formed on one side of the thermoelectric element 530, and the cooling layer 532 may be formed on the opposite side.

[0113] The heating layer 531 may be positioned above the thermoelectric element 530, and the cooling layer 532 may be positioned below the thermoelectric element 530. The heating layer 531 may face the outside of the main body 100 and the cooling layer 532 may face the inside of the first storage compartment 11. The air that has been warmed by heat exchange with the heating layer 531 may be discharged to the outside of the main body 100, and the air that has been cooled by heat exchange with the cooling layer 532 may be supplied to the first storage compartment 11.

[0114] The thermoelectric cooler 400 may include a heat sink 520. The heat sink 520 may be located on the outside of the main body 100. The heat sink 520 may contact the heating layer 531 to absorb heat from the heating layer 531 and release heat to the outside of the main body 100. Heat exchange between the heating layer 531 and the outside air of the main body 100 may be efficiently performed by the heat sink 520. The heat sink 520 may be referred to by various terms, such as a hot sink, a heatsink, and a hot heat sink.

[0115] The heat sink 520 may include a heat sink base 521 that contacts the heating layer 531, and a plurality of heat dissipation fins 525 that protrude from the heat sink base 521 to expand a heat transfer area. The plurality of heat dissipation fins 525 may protrude upward from the heat sink base 521. The heat sink 520 may be formed of a metal material having a high thermal conductivity. For example, the heat sink 520 may be formed of aluminum or copper.

[0116] The thermoelectric cooler 400 may include a cooling sink 570. The cooling sink 570 may be located inside the first storage compartment 11. The cooling sink 570 may absorb heat from the first storage compartment 11 and transfer to the cooling layer 532 of the thermoelectric element 530. Accordingly, the first storage compartment 11 may be cooled. Heat exchange between the cooling layer 532 and the air in the first storage compartment 11 may be efficiently performed by the cooling sink 570. The cooling sink 570 may be referred to by various terms, such as a cold sink, a cold heat sink, and a cooling heat sink.

[0117] The cooling sink 570 may include a cooling sink base 571 that contacts the cooling layer 532 and a plurality

of cooling fins **575** that protrude from the cooling sink base **571** to expand a heat transfer area. The plurality of cooling fins **575** may protrude downward from the cooling sink base **571**. The cooling sink base **571** and the plurality of cooling fins **575** may be formed integrally. The cooling sink **570** may be formed of a metal material having a high thermal conductivity. For example, the cooling sink **570** may be formed of aluminum or copper.

[0118] The heat sink **520** and the cooling sink **570** may allow the heat generation of the heating layer **531** and the heat absorption of the cooling layer **532** to occur smoothly.

[0119] The thermoelectric cooler **400** may include a heat dissipation fan **600** blowing air toward the heat sink **520**. Heat exchange between the heat sink **520** and the outside air of the main body **100** may be efficiently performed by operation of the heat dissipation fan **600**.

[0120] The heat dissipation fan **600** may be positioned in a horizontal direction of the heat sink **520**. The heat dissipation fan **600** may be disposed on the outside of the main body **100**. The heat dissipation fan **600** may be disposed on the upper side of the upper wall **110**. The heat dissipation fan **600** may be a centrifugal fan that draws in air in an axial direction and discharges the air in a radial direction. The centrifugal fan may include a blower fan. A rotation axis **610** of the heat dissipation fan **600** may be disposed vertically on the upper side of the upper wall **110**.

[0121] The thermoelectric cooler **400** may include a heat dissipation duct **700** to guide air flowing by the heat dissipation fan **600**. The heat dissipation duct **700** may be disposed on the upper side of the main body **100**. The heat dissipation fan **600** may be located inside the heat dissipation duct **700**. The heat sink **520** may be located inside the heat dissipation duct **700**. The heat dissipation duct **700** may be disposed on the upper side of the upper wall **110**.

[0122] The heat dissipation duct **700** may guide the outside air of the main body **100** to the heat sink **520**. Air that has exchanged heat with the heat sink **520** may be discharged to the outside of the main body **100** through the heat dissipation duct **700**. The heat dissipation duct **700** may include an outside air inlet **751** for drawing the outside air of the main body **100** into the heat dissipation duct **700**, and an outside air outlet **782** for discharging air that has exchanged heat with the heat sink **520** to the outside of the main body **100**.

[0123] The thermoelectric cooler **400** may include a cooling fan **800** blowing air toward the cooling sink **570**. Heat exchange between the cooling sink **570** and the air inside the first storage compartment **11** may be efficiently performed by operation of the cooling fan **800**.

[0124] The cooling fan **800** may be positioned in a horizontal direction of the cooling sink **570**. The cooling fan **800** may be disposed inside the first storage compartment **11**. The cooling fan **800** may be disposed on the lower side of the upper wall **110**. The cooling fan **800** may be a centrifugal fan that draws in air in the axial direction and discharges the air in a radial direction. A rotation axis **810** of the cooling fan **800** may be disposed vertically on the lower side of the upper wall **110**.

[0125] The thermoelectric cooler **400** may include a second temperature sensor **112** for detecting a temperature of air cooled by the cooling fan **800**. The second temperature sensor **112** may detect a temperature of the cooling sink **570**. The temperature of the cooling sink **570** may indicate a temperature of the air around the cooling sink **570** and/or a

surface temperature of the cooling sink **570**. The second temperature sensor **112** may be provided in the cooling sink **570**. The second temperature sensor **112** may also be provided in a cooling duct **900**.

[0126] The thermoelectric cooler **400** may include the cooling duct **900** to guide air flowing by the cooling fan **800**. The cooling fan **800** may be located inside the cooling duct **900**. The cooling sink **570** may be located inside the cooling duct **900**. The cooling duct **900** may be disposed on the lower side of the upper wall **110**.

[0127] The cooling duct **900** may guide air drawn in from the inside of the first storage compartment **11** to the cooling sink **570**. Air that has exchanged heat with the cooling sink **570** may be discharged back into the first storage compartment **11** through the cooling duct **900**. The cooling duct **900** may include an internal air inlet **991** for drawing air inside the first storage compartment **11** into the cooling duct **900**, and an internal air outlet **992** for discharging the air that has exchanged heat with the cooling sink **570** into the first storage compartment **11**.

[0128] The refrigerator **1** according to the disclosure may supply cold air to the first storage compartment **11** using various methods. The refrigerator **1** may operate at least one of the thermoelectric cooler **400** or the refrigeration cycle device to cool the first storage compartment **11**. For example, the refrigerator **1** may provide a first method of supplying only cold air generated by the thermoelectric cooler **400**, a second method of supplying only cold air generated by the refrigeration cycle device, and a third method of supplying both cold air generated by the thermoelectric cooler **400** and cold air generated by the refrigeration cycle device.

[0129] The refrigerator **1** may supply cold air to the first storage compartment **11** using an appropriate method according to external and internal conditions. For example, the refrigerator **1** may cool the first storage compartment **11** using one of the methods according to an indoor temperature in which the refrigerator **1** is installed. In a case where an indoor temperature is relatively high and cooling by the cooling cycle is more effective than cooling by the thermoelectric cooler **400**, the refrigerator **1** may cool the first storage compartment **11** only with the cold air generated by the refrigeration cycle device. In a case where an indoor temperature is relatively low and cooling performance may be sufficiently achieved only with cooling by the thermoelectric cooler **400**, the refrigerator **1** may cool the first storage compartment **11** only with the cold air generated by the thermoelectric cooler **400**.

[0130] The refrigerator **1** may operate only the thermoelectric cooler **400** in a case where noise reduction is required. In a case where rapid cooling of the first storage compartment **11** is required, the refrigerator **1** may simultaneously supply cold air generated by the thermoelectric cooler **400** and cold air generated by the refrigeration cycle device to the first storage compartment **11**.

[0131] Although it has been described that the refrigerator **1** includes the thermoelectric cooler **400** and the refrigeration cycle device, the disclosure is not limited thereto. The refrigerator **1** may include only one of the thermoelectric cooler **400** and the refrigeration cycle device.

[0132] Although it has been described that the thermoelectric cooler **400** is disposed on the upper wall **110** of the main body **100**, the position of the thermoelectric cooler **400** is not limited thereto. According to various embodiments,

the thermoelectric cooler 400 may be disposed on at least one of the upper wall 110, the lower wall 120, the left wall 130, the right wall 140, or the rear wall 150.

[0133] FIG. 6 is an exploded view of a heat dissipation fan and a thermoelectric module according to an embodiment.

[0134] Referring to FIG. 6, the thermoelectric cooler 400 may include a thermoelectric module 500. The thermoelectric module 500 may include the heat sink 520, the thermoelectric element 530, a module plate 550, and the cooling sink 570. The heat sink 520, the thermoelectric element 530, and the cooling sink 570 may be assembled or formed integrally.

[0135] The module plate 550 may serve as a frame of the thermoelectric module 500. The module plate 550 may be formed of a resin material having a low thermal conductivity. The module plate 550 may maintain a gap between the heat sink 520 and the cooling sink 570, and may support the heat sink 520 and the cooling sink 570. The module plate 550 may be formed integrally with a fan case 650. However, the module plate 550 may be provided separately from the fan case 650.

[0136] The module plate 550 may include a heat sink support 552 supporting the heat sink 520.

[0137] The module plate 550 may include a module plate opening 551. The thermoelectric element 530 may be disposed inside the module plate opening 551. A vertical length of the module plate opening 551 may be larger than that of the thermoelectric element 530, and the thermoelectric element 530 may be disposed at an upper end of the module plate opening 551. In general, because the heat generation amount of the thermoelectric element 530 is higher than the heat absorption amount, it is advantageous for heat dissipation of the heating layer 531 that the thermoelectric element 530 is located at the upper end of the module plate opening 551.

[0138] The cooling sink 570 may include a cooling conductive portion 574 protruding from the cooling sink base 571. The cooling conductive portion 574 may be in contact with the cooling layer 532 of the thermoelectric element 530.

[0139] The thermoelectric module 500 may include an element insulation 540 to insulate the module plate 550 and the thermoelectric element 530. The element insulation 540 may be disposed in the module plate opening 551 to prevent sides of the thermoelectric element 530 from contacting the module plate 550. The element insulation 540 may include an element insulation opening 541, and the thermoelectric element 530 may be accommodated in the element insulation opening 541.

[0140] The thermoelectric module 500 may include a sink insulation 580 between the module plate 550 and the cooling sink 570. The sink insulation 580 may prevent heat from being transferred between the heat sink 520 and the cooling sink 570 through the module plate 550. The sink insulation 580 may include a sink insulation opening 581.

[0141] The sink insulation 580 may be omitted. In a case where the sink insulation 580 is omitted, the heat sink 520 may be supported on an upper side of the module plate 550 and the cooling sink 570 may be supported on a lower side of the module plate 550.

[0142] The heat dissipation fan 600 may be installed in the fan case 650. The fan case 650 may form a flow path of air

flowing by the heat dissipation fan 600. The fan case 650 may be formed integrally with the module plate 550 or may be provided separately.

[0143] The fan case 650 may include a case bottom 650 on which the heat dissipation fan 600 is rotatably installed, and a case scroll portion 670 extending upward from an edge of the case bottom 650 to guide air blown from the heat dissipation fan 600 toward the heat sink 520. The heat dissipation fan 600 may be a centrifugal fan, and may be installed on the case bottom 650 to allow the rotation axis 610 to be perpendicular to the case bottom 650. In addition, the heat sink 520 may be positioned in a radial direction of the heat dissipation fan 600. With the structure described above, an overall vertical length of the thermoelectric cooler 400 may be compact.

[0144] The case scroll portion 670 may be formed to surround the heat dissipation fan 600. The case scroll portion 670 may have a scroll portion opening 673 open toward the heat sink 520. The case scroll portion 670 may include a downstream end 671 along a rotational direction R of the heat dissipation fan 600 and an upstream end 672 along the rotational direction R.

[0145] The fan case 650 may include a case guide 680 to guide air flowing from the heat dissipation fan 600 to the vicinity of the downstream end 671 of the case scroll portion 670.

[0146] FIG. 7 illustrates a heat sink according to an embodiment.

[0147] Referring to FIG. 7, a plurality of heat dissipation fins 525 may protrude from an upper side 522 of the heat sink base 521. The plurality of heat dissipation fins 525 may protrude in a first direction 526 perpendicular to the upper side 522 of the heat sink base 521.

[0148] The plurality of heat dissipation fins 525 may be formed to extend in a second direction 527 parallel to the upper side 522 of the heat sink base 521. The second direction 527 may be perpendicular to the first direction 526. A first heat dissipation channel 528 may be formed between two adjacent heat dissipation fins 525. The first heat dissipation channel 528 may extend in the second direction 527 like the plurality of heat dissipation fins 525.

[0149] A second heat dissipation channel 529 having a different width from the first heat dissipation channel 528 may be formed between two specific heat dissipation fins 525 among the plurality of heat dissipation fins 525. A width of the second heat dissipation channel 529 may be larger than that of the first heat dissipation channel 528.

[0150] Air flowing by an operation of the heat dissipation fan 600 may pass through the heat dissipation channels 528 and 529, and may exchange heat with the plurality of heat dissipation fins 525. An air flow A flowing by the operation of the heat dissipation fan 600 may pass through the heat dissipation channels 528 and 529 in a direction parallel to the second direction 527.

[0151] FIG. 8 illustrates a cooling sink according to an embodiment.

[0152] Referring to FIG. 8, a plurality of cooling fins 575 may protrude from a lower side 572 of the cooling sink base 571. The plurality of cooling fins 575 may protrude in a first direction 576 perpendicular to the lower side 572 of the cooling sink base 571.

[0153] The plurality of cooling fins 575 may be formed to extend in a second direction 577 parallel to the lower side 572 of the cooling sink base 571. The second direction 577

may be perpendicular to the first direction 576. A first cooling channel 578 may be formed between two adjacent cooling fins 575.

[0154] A second cooling channel 579 having a different width from the first cooling channel 578 may be formed between two specific cooling fins 575 among the plurality of cooling fins 575. A width of the second cooling channel 579 may be larger than that of the first cooling channel 578.

[0155] Air flowing by an operation of the cooling fan 800 may pass through the cooling channels 578 and 579 and may exchange heat with the plurality of cooling fins 575. An air flow B flowing by the operation of the cooling fan 800 may pass through the cooling channels 578 and 579 in a direction parallel to the second direction 577.

[0156] FIG. 9 is a view illustrating a state in which a cooling duct cover and a cooling duct body are separated from an upper wall of a main body according to an embodiment. FIG. 10 is an exploded view of a cooling duct cover and a cooling duct body, viewed from below, according to an embodiment. FIG. 11 is an exploded view of a cooling duct cover and a cooling duct body, viewed from above, according to an embodiment.

[0157] Referring to FIG. 9, FIG. 10 and FIG. 11, the refrigerator 1 may include the cooling duct 900 to guide cold air generated by exchanging heat with the cooling sink 570 to the first storage compartment 11. The cooling duct 900 may be located on an upper part of the first storage compartment 11. The cooling duct 900 may be located on the lower side of the upper wall 110.

[0158] The cooling duct 900 may include a cooling duct body 910 coupled to the upper wall 110 of the main body 100, and a cooling duct cover 940 coupled to a lower part of the cooling duct body 910. A cooling space 990 may be formed by the cooling duct body 910 and the cooling duct cover 940. The cooling sink 570 may be disposed in the cooling space 990.

[0159] The cooling duct 900 may be coupled to a connecting frame 210. An elastic coupling protrusion 950 protruding upward may be formed on the cooling duct cover 940. A coupling protrusion through-hole 913 through which the elastic coupling protrusion 950 penetrates may be formed in the cooling duct body 910. The elastic coupling protrusion 950 may penetrate the coupling protrusion through-hole 913 and be coupled to the connecting frame 210.

[0160] The cooling duct 900 may be coupled to the connecting frame 210 through a separate cooling duct coupling member 901. The cooling duct coupling member 901 may be a screw. A coupling hole 951 through which the cooling duct coupling member 901 is coupled may be formed in the cooling duct cover 940. A coupling hole 912 through which the cooling duct coupling member 901 is coupled may be formed in the cooling duct body 910.

[0161] An example in which the cooling duct 900 is coupled to the connecting frame 210 has been described, but the disclosure is not limited thereto. For example, the cooling duct 900 may be coupled to the upper wall 110 instead of the connecting frame 210.

[0162] The cooling duct body 910 may include a sink through hole 911 through which the cooling sink 570 passes. The cooling sink 570 may pass through the sink through hole 911 and be disposed in the cooling space 990. The cooling duct body 910 may include a fan installation portion 920 protruding upward. A fan accommodating space 921 in

which the cooling fan 800 is accommodated may be formed on a lower side of the fan installation portion 920.

[0163] The cooling fan 800 may be a centrifugal fan and be installed on the lower side of the fan installation portion 920 of the cooling duct body 910, and the rotation axis 810 of the cooling fan 800 may be perpendicular to the lower side of the fan installation portion 920. The cooling sink 570 may be positioned in a radial direction of the cooling fan 800. With the structure described above, a vertical length of the cooling duct 900 may be compact.

[0164] The internal air inlet 991 may be formed on a bottom 941 of the cooling duct cover 940. Air in the first storage compartment 11 may be drawn into the cooling duct 900 through the internal air inlet 991. The internal air inlet 991 may be formed below the cooling fan 800. The cooling duct 900 may include the internal air outlet 992. Air that has exchanged heat with the cooling sink 570 may be discharged into the first storage compartment 11 through the internal air outlet 992. The internal air outlet 992 may be formed at a front part of the cooling duct 900. The internal air outlet 992 may be formed between a front part of the cooling duct body 910 and a front part of the cooling duct cover 940. The internal air outlet 992 may be elongated along the left-right longitudinal direction of the cooling duct 900.

[0165] Because the internal air outlet 992 is formed at the front part of the cooling duct 900, the air discharged through the internal air outlet 992 may be blown to the front of the first storage compartment 11. Accordingly, an air curtain in which air flows vertically downward to the front of the first storage compartment 11 may be formed by the air discharged through the internal air outlet 992.

[0166] The internal air outlet 992 may be located close to the front of the first storage compartment 11. Compared to the illustrated example, the cooling duct 900 itself may be located closer to the front of the first storage compartment 11, or a length of the cooling duct 900 in the front-back direction may be formed long to allow the internal air outlet 992 to be located closer to the front of the first storage compartment 11. Such structures may form the air curtain more efficiently.

[0167] Frost formed on the surface of the cooling sink 570 may melt and generate defrost water. The defrost water may drip from the cooling sink 570 onto the bottom 941 of the cooling duct cover 940. The cooling duct 900 may accommodate the defrost water generated during the defrosting of the cooling sink 570. In other words, the cooling duct 900 may store the defrost water.

[0168] The cooling duct cover 940 may include a defrost water guide 980 protruding from the bottom 941 of the cooling duct cover 940. The defrost water guide 980 may be formed to extend in the front-back direction. The defrost water guide 980 may contain the defrost water. The defrost water guide 980 may form a space having an area corresponding to the area of the cooling sink 570. A gap may be formed between the defrost water guide 980 and the cooling sink 570.

[0169] FIG. 12 is a control block diagram of a refrigerator according to an embodiment.

[0170] Referring to FIG. 12, the refrigerator 1 according to an embodiment may include a compressor 2, a door sensor 25, the evaporator fan 80, the first temperature sensor 111, the second temperature sensor 112, the thermoelectric element 530, the heat dissipation fan 600, the cooling fan 800, and a controller 1000. The controller 1000 may be electri-

cally connected to various electronic components of the refrigerator 1 and may control each electronic component.

[0171] The controller 1000 may include at least one processor 1020 controlling the operation of the refrigerator 1, and at least one memory 1010 storing programs, software, instructions, and/or data for controlling the operation of the refrigerator 1.

[0172] The memory 1010 may store various programs, software, instructions, and/or data required for the operation of the refrigerator 1. The memory 1010 may be implemented as a memory embedded in the refrigerator 1 or as a memory detachable from the refrigerator 1 depending on a data storage use.

[0173] For example, data for driving the refrigerator 1 may be stored in the memory embedded in the refrigerator 1. Data for an extended function of the refrigerator 1 may be stored in the memory that may be inserted into the refrigerator 1.

[0174] The memory embedded in the refrigerator 1 may be implemented as at least one of a volatile memory or a non-volatile memory. For example, the volatile memory may include a dynamic random access memory (DRAM), static RAM (SRAM), and/or synchronous dynamic RAM (SDRAM). The non-volatile memory may include an one time programmable read only memory (OTPROM), programmable ROM (PROM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), mask ROM, flash ROM, flash memory, a hard drive, and/or a solid state drive (SSD).

[0175] The memory that may be inserted into the refrigerator 1 may include a memory card such as a compact flash (CF), a secure digital (SD), a micro secure digital (Micro-SD), a mini secure digital (Mini-SD), an extreme digital (xD), or a multi-media card (MMC). In addition, a universal serial bus (USB) memory device that may be connected to a USB port may also be inserted into the refrigerator 1.

[0176] The processor 1020 may control an operation of the refrigerator 1. The processor 1020 may control various components of the refrigerator 1. For example, the processor 1020 may control the compressor 2, the door sensor 25, the evaporator fan 80, the first temperature sensor 111, the second temperature sensor 112, the thermoelectric element 530, the heat dissipation fan 600, and the cooling fan 800.

[0177] The processor 1020 may perform various operations of the refrigerator 1 by processing the programs, software, instructions, and/or data stored in the memory 1010.

[0178] The processor 1020 may include at least one of a central processing unit (CPU), graphics processing unit (GPU), accelerated processing unit (APU), many integrated core (MIC), digital signal processor (DSP), neural processing unit (NPU), hardware accelerator, or machine learning accelerator.

[0179] The compressor 2 may compress the refrigerant and supply the compressed refrigerant to a heat exchanger (e.g., the condenser and the evaporator 3). The processor 1020 may control the compressor 2 to adjust a temperature of the cold air generated in the evaporator 3. The processor 1020 may control the compressor 2 to maintain temperatures of the storage compartments 11, 12, and 13 at a target temperature. The processor 1020 may adjust an operating frequency of the compressor 2. The processor 1020 may control the on/off of the compressor 2.

[0180] The door sensor 25 may detect opening and/or closing of the doors 21, 22, 23, and 24. The door sensor 25 may be disposed for each of the plurality of doors. The door sensor 25 may transmit an opening signal corresponding to opening of the door to the processor 1020. The door sensor 25 may transmit a closing signal corresponding to closing of the doors 21, 22, 23, and 24 to the processor 1020. The processor 1020 may identify the opening and closing of the door based on the signal transmitted from the doors 21, 22, 23, and 24.

[0181] The evaporator fan 80 causes air to flow. The evaporator fan 80 may move the air cooled by the evaporator 3 to the storage compartments 11, 12, and 13. The processor 1020 may control a rotation speed of the evaporator fan 80. The processor 1020 may control an input voltage of the evaporator fan 80 to control the rotation speed of the evaporator fan 80. As the input voltage of the evaporator fan 80 increases, the rotation speed of the evaporator fan 80 may increase. As the rotation speed of the evaporator fan 80 increases, a flow rate of air flowing into the storage compartments 11, 12, and 13 may increase. Conversely, as the rotation speed of the evaporator fan 80 decreases, the flow rate of air flowing into the storage compartments 11, 12, and 13 may decrease.

[0182] The first temperature sensor 111 may detect a temperature of the evaporator 3. The first temperature sensor 111 may transmit an electrical signal, data, and/or information corresponding to the temperature of the evaporator 3 to the controller 1000. The processor 1020 may identify the temperature of the evaporator 3 based on the electrical signal, data, and/or information transmitted from the first temperature sensor 111.

[0183] The second temperature sensor 112 may detect a temperature of the cooling sink 570. The second temperature sensor 112 may transmit an electrical signal, data, and/or information corresponding to the temperature of the cooling sink 570 to the controller 1000. The processor 1020 may identify the temperature of the cooling sink 570 based on the electrical signal, data, and/or information transmitted from the second temperature sensor 112.

[0184] The refrigerator 1 may include various sensors in addition to the sensors described above. For example, the refrigerator 1 may include a third temperature sensor for detecting a temperature of the heat sink 520. The refrigerator 1 may include a fourth temperature sensor for detecting temperatures of the storage compartments 11, 12, and 13. In addition, the refrigerator 1 may include a humidity sensor for detecting a humidity of the first storage compartment 11.

[0185] The thermoelectric element 530 may be controlled by the processor 1020. The processor 1020 may control the on/off of the thermoelectric element 530. Turning the thermoelectric element 530 on may indicate supplying power to the thermoelectric element 530. Turning the thermoelectric element 530 off may indicate stopping the supply of power to the thermoelectric element 530. The processor 1020 may adjust at least one of the voltage or current applied to the thermoelectric element 530.

[0186] As power is supplied to the thermoelectric element 530, a heat generation process may occur in the heating layer 531 and a heat absorption process may occur in the cooling layer 532. Air may be heated by the heat sink 520 in contact with the heating layer 531. The air heated by the heat sink 520 may be discharged to the outside of the main body 100. Air may be cooled by the cooling sink 570 in contact with

the cooling layer 532. The air cooled by the cooling sink 570 may be supplied to the first storage compartment 11.

[0187] The processor 1020 may adjust a duty ratio of a voltage pulse applied to the thermoelectric element 530. The voltage pulse may be applied to the thermoelectric element 530 at defined time intervals (periods). Within a single period, the voltage pulse consists of an on-duty and an off-duty. The duty ratio may represent the proportion of on-duty within a single period. An increase in the duty ratio of the voltage pulses indicates an increase in the on-duty of the voltage pulse. An increase in the duty ratio of the voltage pulse applied to the thermoelectric element 530 may increase a cooling performance of the thermoelectric element 530. The increase in the cooling performance of the thermoelectric element 530 may decrease a temperature of the first storage compartment 11. A decrease in the duty ratio of the voltage pulse applied to the thermoelectric element 530 may decrease the cooling performance of the thermoelectric element 530. The decrease in the cooling performance of the thermoelectric element 530 may increase the temperature of the first storage compartment 11.

[0188] The heat dissipation fan 600 may supply air outside the main body 100 to the heat sink 520, and may discharge the air that has exchanged heat with the heat sink 520 to the outside of the main body 100. The air that has exchanged heat with the heat sink 520 may move quickly according to the operation of the heat dissipation fan 600, and a heat dissipation performance of the heat sink 520 may increase.

[0189] The cooling fan 800 may draw in air from the first storage compartment 11, supply the air to the cooling sink 570, and discharge the air that has exchanged heat with the cooling sink 570 into the first storage compartment 11. The air that has exchanged heat with the cooling sink 570 may move quickly according to the operation of the cooling fan 800, and the first storage compartment 11 may be cooled quickly.

[0190] The processor 1020 may control the on/off of each of the heat dissipation fan 600 and the cooling fan 800. The heat dissipation fan 600 and the cooling fan 800 may be controlled independently. The processor 1020 may adjust a rotation speed of each of the heat dissipation fan 600 and the cooling fan 800. The processor 1020 may adjust an input voltage of each of the heat dissipation fan 600 and the cooling fan 800 to adjust the rotation speed of each of the heat dissipation fan 600 and the cooling fan 800. An increase in the input voltages of the heat dissipation fan 600 and the cooling fan 800 may cause the rotation speeds of the heat dissipation fan 600 and the cooling fan 800 to increase. A decrease in the input voltages of the heat dissipation fan 600 and the cooling fan 800 may cause the rotation speeds of the heat dissipation fan 600 and the cooling fan 800 to decrease. The heat dissipation fan 600 and the cooling fan 800 may include a brushless direct current (DC) motor.

[0191] The operation of the thermoelectric element 530 and the operation of each of the cooling fan 800 and the heat dissipation fan 600 may or may not be linked. For example, the processor 1020 may operate the cooling fan 800 and the heat dissipation fan 600 simultaneously with the operation of the thermoelectric element 530. The processor 1020 may operate the cooling fan 800 and the heat dissipation fan 600 after operating the thermoelectric element 530. The processor 1020 may also operate the thermoelectric element 530 after operating the cooling fan 800 and the heat dissipation fan 600.

[0192] The processor 1020 may stop the operation of the cooling fan 800 and the heat dissipation fan 600 at the same time as stopping the operation of the thermoelectric element 530. The processor 1020 may stop the operation of the cooling fan 800 and the heat dissipation fan 600 after stopping the operation of the thermoelectric element 530. The processor 1020 may stop the operation of the thermoelectric element 530 after stopping the operation of the cooling fan 800 and the heat dissipation fan 600. In addition, in a case where the cooling sink 570 requires to be defrosted, the processor 1020 may stop the operation of the thermoelectric element 530 and operate the cooling fan 800.

[0193] In a case where the thermoelectric element 530 is stopped during operation, the cooling fan 800 may be operated for a defined drying time and then be stopped in order to dry the cooling fan 800.

[0194] In addition, the processor 1020 may control the operation of each of the heat dissipation fan 600 and the cooling fan 800 based on an operating state of the compressor 2. For example, based on the operation of the compressor 2 being stopped, the processor 1020 may stop the operation of each of the heat dissipation fan 600 and the cooling fan 800, or may reduce the rotation speed of each of the heat dissipation fan 600 and the cooling fan 800.

[0195] The processor 1020 may stop the operation of the compressor 2 based on completion of cooling of the first storage compartment 11 or an occurrence of an error in the refrigeration cycle device. For example, the processor 1020 may determine the completion of cooling of the first storage compartment 11 based on a temperature of the first storage compartment 11 reaching a predefined target temperature. In addition, the processor 1020 may determine that an error has occurred in the refrigeration cycle device in the event of a failure of the compressor 2, the expansion valve, and/or the first temperature sensor 111.

[0196] The refrigerator 1 may reduce noise by stopping the operation of each of the heat dissipation fan 600 and the cooling fan 800 or by reducing the rotation speed of each of the heat dissipation fan 600 and the cooling fan 800 while the operation of the compressor 2 is stopped.

[0197] As described above, the refrigerator 1 may operate at least one of the thermoelectric cooler 400 or the refrigeration cycle device to cool the first storage compartment 11. For example, the processor 1020 may turn off the compressor 2 and turn on the thermoelectric element 530, thereby cooling the first storage compartment 11 using the thermoelectric cooler 400. The processor 1020 may turn on the compressor 2 and turn off the thermoelectric element 530, thereby cooling the first storage compartment 11 using the refrigeration cycle device. The processor 1020 may turn on both the compressor 2 and the thermoelectric element 530, thereby cooling the first storage compartment 11 using both the thermoelectric cooler 400 and the refrigeration cycle device.

[0198] In a case where the thermoelectric cooler 400 operates to cool the first storage compartment 11, frost may form on the surfaces of the cooling sink 570. When frost is formed on the cooling sink 570, a heat exchange efficiency between the cooling sink 570 and the air decreases, and a cooling performance of the thermoelectric cooler 400 decreases. Accordingly, in order to prevent the cooling performance of the thermoelectric cooler 400 from decreasing, a defrosting operation for the cooling sink 570 may be required.

[0199] The processor 1020 may perform the defrosting operation for the cooling sink 570 based on a defined defrosting condition being satisfied. For example, the processor 1020 may perform the defrosting operation for the cooling sink 570 based on a temperature of the cooling sink 570 being lower than or equal to a defined freezing temperature. In addition, the processor 1020 may perform the defrosting operation for the cooling sink 570 at defined time intervals. The defrosting conditions of the cooling sink 570 are not limited to the above examples. The defrosting conditions of the cooling sink 570 may be provided in various ways depending on the design.

[0200] The processor 1020 may perform the defrosting operation for the cooling sink 570 by stopping the operation of the thermoelectric element 530. Stopping the operation of the thermoelectric element 530 may refer to stopping the power supply to the thermoelectric element 530. In addition, the processor 1020 may perform the defrosting operation for the cooling sink 570 by stopping the operation of the thermoelectric element 530 and operating the heat dissipation fan 600 and the cooling fan 800.

[0201] The processor 1020 may terminate the defrosting operation for the cooling sink 570 based on a defined defrosting termination condition being satisfied. For example, the processor 1020 may terminate the defrosting operation for the cooling sink 570 based on a temperature of the cooling sink 570 being greater than or equal to a melting point of the frost. The processor 1020 may operate (i.e., turn on) the thermoelectric element 530 based on determining the termination of the defrosting operation. The defrosting termination condition of the cooling sink 570 is not limited to the above example. The defrosting termination condition of the cooling sink 570 may be provided in various ways depending on the design.

[0202] When the operation of the thermoelectric element 530 is stopped, the temperature of the cooling sink 570 increases, causing the frost to melt. The defrost water generated by the melting of the frost may be accommodated in the cooling duct 900. The processor 1020 may operate the cooling fan 800 to evaporate the defrost water in the cooling duct 900 according to the defrosting of the cooling sink 570.

[0203] As the cooling fan 800 operates, the defrost water may evaporate. The refrigerator 1 may remove the defrost water by evaporating the defrost water accommodated in the cooling duct 900. The defrost water evaporated in the cooling duct 900 may flow into the first storage compartment 11. The evaporated defrost water may move to the evaporator 3 through the first storage compartment 11 and the return duct 73. The evaporated defrost water may be condensed in the evaporator 3 and discharged to the machine compartment. The defrost water discharged to the machine compartment may be evaporated again in the machine compartment and discharged to the outside.

[0204] In order to prevent the evaporated defrost water from being drawn back into the cooling duct 900 and to move the evaporated defrost water more quickly to the evaporator 3, the operation of the cooling fan 800 may be temporarily stopped. The evaporated defrost water may be moved to the evaporator 3 by an operation of the evaporator fan 80. For example, based on an elapse of a defined evaporation time of the defrost water, the processor 1020 may temporarily stop the operation of the cooling fan 800 to allow the moisture in the first storage compartment 11 to

move to the evaporator 3 through the return duct 73 according to the operation of the evaporator fan 80.

[0205] Meanwhile, in a case where the doors 21, 22, 23, and 24 are open for a long time, external moisture may enter the storage compartments 11, 12, and 13. External moisture may condense in the cooling sink 570 and be stored in the cooling duct 900, which may cause water overflow in the cooling duct 900. The processor 1020 of the refrigerator 1 may stop the operation of the thermoelectric element 530 and the cooling fan 800 based on the doors 21, 22, 23, and 24 of the refrigerator 1 being opened for a defined threshold time (e.g., 10 minutes) to prevent water overflow in the cooling duct 900. By stopping the operation of the thermoelectric element 530 and the cooling fan 800, external moisture may be prevented from condensing in the cooling sink 570.

[0206] In a case where an error occurs in the refrigeration cycle device and the operation of the compressor 2 is stopped, the defrost water may not be condensed in the evaporator 3. Because the evaporated defrost water is not condensed in the evaporator 3, the defrost water may flow back to the first storage compartment 11 and be accommodated in the cooling duct 900 again. In other words, even when an error occurs in the refrigeration cycle device, an overflow of water may occur in the cooling duct 900. In the case of the error in the refrigeration cycle device, the processor 1020 may stop the operation of the thermoelectric element 530 and the cooling fan 800.

[0207] As the thermoelectric element 530 operates and the air is cooled in the cooling sink 570, the defrost water accommodated in the cooling duct 900 may freeze. As the defrost water freezes, the air in the cooling duct 900 may not move smoothly, and thus the defrost water is required to be prevented from freezing.

[0208] The processor 1020 may adjust the duty ratio of the voltage pulse applied to the thermoelectric element 530 based on the temperature of the cooling sink 570, thereby performing anti-freezing control for the defrost water. The processor 1020 may determine the duty ratio of the voltage pulse applied to the thermoelectric element 530 based on the temperature of the cooling sink 570 at defined time intervals (e.g., 10 minutes) in order to maintain the temperature of the cooling sink 570 within a temperature range which is preset to prevent the defrost water from freezing.

[0209] For example, the processor 1020 may reduce the duty ratio of the voltage pulse based on the temperature of the cooling sink 570 being lower than or equal to a first threshold temperature (e.g., -1.5 degrees Celsius). The processor 1020 may increase the duty ratio of the voltage pulse based on the temperature of the cooling sink 570 being higher than or equal to a second threshold temperature (e.g., 0.5 degrees Celsius) that is higher than the first threshold temperature. The processor 1020 may maintain the duty ratio of the voltage pulse based on the temperature of the cooling sink 570 being higher than the first threshold temperature and lower than the second threshold temperature.

[0210] In addition, the processor 1020 may temporarily stop the anti-freezing control for the defrost water based on a priority operation condition of the thermoelectric element 530 being satisfied. The priority operation condition of the thermoelectric element 530 may be to increase a cooling performance of the thermoelectric element 530. For example, in a case where an operation mode of the refrigerator 1 is set to a power cooling mode, in a case where the

operation mode of the refrigerator **1** is set to a test mode for forcibly starting the thermoelectric element **530**, or in a case where an external temperature of the refrigerator **1** is higher than or equal to a defined limit temperature, it may be determined that the priority operation condition of the thermoelectric element **530** is satisfied. The priority operation condition of the thermoelectric element **530** is not limited to the above examples. The priority operation condition of the thermoelectric element **530** may be provided in various ways depending on the design.

[0211] As such, the refrigerator **1** according to the disclosure may remove the defrost water by evaporating the defrost water accommodated in the cooling duct **900**. That is, the refrigerator **1** does not require a complex structure for removing the defrost water generated in the cooling sink **570**. In addition, the refrigerator **1** may prevent the defrost water in the cooling duct **900** from freezing and the water in the cooling duct **900** from overflowing, thereby preventing a performance of the thermoelectric cooler **400** from deteriorating.

[0212] FIG. **13** is a flowchart briefly illustrating a method for controlling a refrigerator according to an embodiment.

[0213] Referring to FIG. **13**, the processor **1020** of the refrigerator **1** may perform a defrosting operation for the cooling sink **570** (**1301**). The processor **1020** may perform the defrosting operation for the cooling sink **570** based on a defined defrosting condition being satisfied. For example, the processor **1020** may perform the defrosting operation for the cooling sink **570** based on a temperature of the cooling sink **570** being lower than or equal to a defined freezing temperature. In addition, the processor **1020** may perform the defrosting operation for the cooling sink **570** at defined time intervals.

[0214] The processor **1020** may perform the defrosting operation for the cooling sink **570** by stopping (i.e., turning off) an operation of the thermoelectric element **530**. In addition, the processor **1020** may perform the defrosting operation for the cooling sink **570** by stopping the operation of the thermoelectric element **530** and operating the heat dissipation fan **600** and the cooling fan **800**. When the operation of the thermoelectric element **530** is stopped, the temperature of the cooling sink **570** increases, causing the frost to melt. The defrost water generated by the melting of the frost may be accommodated in the cooling duct **900**.

[0215] The processor **1020** may perform an evaporation operation for the defrost water to remove the defrost water accommodated in the cooling duct **900** (**1302**). The processor **1020** may operate the cooling fan **800** to evaporate the defrost water accommodated in the cooling duct **900**. In a case where the cooling fan **800** operates during the defrosting operation for the cooling sink **570**, the defrosting of the cooling sink **570** and the evaporation of the defrost water may be performed simultaneously. In a case where the cooling fan **800** does not operate during the defrosting operation for the cooling sink **570**, the evaporation operation for the defrost water may be performed after the defrosting operation for the cooling sink **570** is completed.

[0216] The defrost water evaporated in the cooling duct **900** may flow into the first storage compartment **11**. The evaporated defrost water may move to the evaporator **3** through the first storage compartment **11** and the return duct **73**. The evaporated defrost water may be condensed in the evaporator **3** and discharged to the machine compartment.

The defrost water discharged to the machine compartment may be evaporated again in the machine compartment and discharged to the outside.

[0217] In order to prevent the defrost water in the cooling duct **900** from freezing, the processor **1020** may perform anti-freezing control for the defrost water (**1303**). As the thermoelectric element **530** operates and the air is cooled in the cooling sink **570**, the defrost water accommodated in the cooling duct **900** may freeze. As the defrost water freezes, the air in the cooling duct **900** may not move smoothly, and thus the defrost water is required to be prevented from freezing.

[0218] The processor **1020** may adjust a duty ratio of a voltage pulse applied to the thermoelectric element **530** based on the temperature of the cooling sink **570**, thereby performing anti-freezing control for the defrost water. The processor **1020** may determine the duty ratio of the voltage pulse applied to the thermoelectric element **530** based on the temperature of the cooling sink **570** at defined time intervals (e.g., 10 minutes) in order to maintain the temperature of the cooling sink **570** within a temperature range which is preset to prevent the defrost water from freezing.

[0219] FIG. **14** is a flowchart illustrating a method for controlling a refrigerator according to an embodiment in greater detail.

[0220] Referring to FIG. **14**, the processor **1020** of the refrigerator **1** may operate the cooling fan **800** of the thermoelectric cooler **400** to evaporate the defrost water in the cooling duct **900** (**1401**).

[0221] The processor **1020** may determine whether a water overflow condition of the cooling duct **900** is satisfied (**1402**). For example, the processor **1020** may determine that the water overflow condition of the cooling duct **900** is satisfied based on the doors **21**, **22**, **23**, and **24** of the refrigerator **1** being continuously opened for a defined threshold time. In addition, the processor **1020** may determine that the water overflow condition of the cooling duct **900** is satisfied based on an occurrence of an error in the refrigeration cycle device (e.g., a failure of the compressor **2**). The processor **1020** may stop the operation of the thermoelectric element **530** and the cooling fan **800** (**1403**), based on the water overflow condition of the cooling duct **900** being satisfied. Based on the water overflow condition of the cooling duct **900** not being satisfied, the anti-freezing control for the defrost water may be performed.

[0222] The processor **1020** may detect a temperature of the cooling sink **570** by controlling the second temperature sensor **112** (**1404**). The temperature of the cooling sink **570** may be detected at defined time intervals (e.g., 10 minutes).

[0223] The processor **1020** may identify whether the temperature of the cooling sink **570** is higher than a first threshold temperature (e.g., -1.5 degrees Celsius) (**1405**). The processor **1020** may reduce the duty ratio of the voltage pulse (**1406**), based on the temperature of the cooling sink **570** being lower than or equal to the first threshold temperature.

[0224] In a case where the temperature of the cooling sink **570** is higher than the first threshold temperature, the processor **1020** may identify whether the temperature of the cooling sink **570** is lower than a second threshold temperature (e.g., 0.5 degrees Celsius) (**1407**). The first threshold temperature may be set to be lower than the second threshold temperature. The processor **1020** may increase the duty ratio

of the voltage pulse (1408), based on the temperature of the cooling sink 570 being higher than or equal to the second threshold temperature.

[0225] The processor 1020 may maintain the duty ratio of the voltage pulse applied to the thermoelectric element 530 (1409), based on the temperature of the cooling sink 570 being higher than the first threshold temperature and lower than the second threshold temperature.

[0226] FIG. 15 is a flowchart illustrating a method of temporarily stopping the anti-freezing control for defrost water described in FIG. 13.

[0227] Referring to FIG. 15, the processor 1020 of the refrigerator 1 may determine whether a priority operation condition of the thermoelectric element 530 is satisfied (1501), while performing the anti-freezing control for the defrost water (1303). The processor 1020 may temporarily stop the anti-freezing control for the defrost water (1502), based on the priority operation condition of the thermoelectric element 530 being satisfied. In a case where the priority operation condition of the thermoelectric element 530 is not satisfied or the priority operation condition is released, the processor 1020 may continue to perform the anti-freezing control for the defrost water.

[0228] The priority operation condition of the thermoelectric element 530 may be for increasing a cooling performance of the thermoelectric element 530. For example, in a case where an operation mode of the refrigerator 1 is set to a power cooling mode, in a case where the operation mode of the refrigerator 1 is set to a test mode for forcibly starting the thermoelectric element 530, or in a case where an external temperature of the refrigerator 1 is higher than or equal to a defined limit temperature, it may be determined that the priority operation condition of the thermoelectric element 530 is satisfied. The priority operation condition of the thermoelectric element 530 is not limited to the above examples. The priority operation condition of the thermoelectric element 530 may be provided in various ways depending on the design.

[0229] FIG. 16 is a flowchart further illustrating the control of the cooling fan in the evaporation operation for defrost water described in FIG. 13.

[0230] Referring to FIG. 16, the processor 1020 of the refrigerator 1 may operate the cooling fan 800 of the thermoelectric cooler 400 to evaporate the defrost water accommodated in the cooling duct 900 (1601). The processor 1020 may count time during which the cooling fan 800 operates to evaporate the defrost water of the cooling duct 900 (1602). For example, in a case where the cooling fan 800 operates during the defrosting operation for the cooling sink 570, an operation time of the cooling fan 800 may be counted from the time when the defrosting operation for the cooling sink 570 is started. In a case where the cooling fan 800 does not operate during the defrosting operation for the cooling sink 570, the operation time of the cooling fan 800 may be counted from the time when the cooling fan 800 starts operating after the defrosting operation for the cooling sink 570 is completed.

[0231] The processor 1020 may temporarily stop the operation of the cooling fan 800 (1603), based on an elapse of a defined evaporation time of the defrost water after the start of the operation of the cooling fan 800 for evaporating the defrost water. The operation of the cooling fan 800 may be temporarily stopped to prevent the evaporated defrost water from being drawn back into the cooling duct 900 and

to move the defrost water more quickly to the evaporator 3. The evaporated defrost water may move to the evaporator 3 according to the operation of the evaporator fan 80. For example, the processor 1020 may temporarily stop the operation of the cooling fan 800 to allow the moisture in the first storage compartment 11 to move to the evaporator 3 through the return duct 73 according to the operation of the evaporator fan 80.

[0232] FIG. 17 is a flowchart illustrating a detailed embodiment of the method for controlling the refrigerator described in FIG. 14.

[0233] Referring to FIG. 17, the processor 1020 of the refrigerator 1 may identify whether an error has occurred in the refrigeration cycle device (1701). In a case where an error has occurred in the refrigeration cycle device, the processor 1020 may turn off the thermoelectric element 530 (1702). In a case where the refrigeration cycle device is operating normally, the processor 1020 may identify whether the doors 21 and 22 of the refrigerating compartment 11 were open for 10 minutes or more (1703). In a case where the doors 21 and 22 of the refrigerating compartment 11 were open for 10 minutes or more, the processor 1020 may turn off the thermoelectric element 530 (1702).

[0234] In a case where the refrigeration cycle device is operating normally and the doors 21 and 22 of the refrigerating compartment 11 are not opened for more than 10 minutes, the processor 1020 may identify whether a temperature of the cooling sink 570 is greater than or equal to 0.5 degrees Celsius (1704). In a case where the temperature of the cooling sink 570 is greater than or equal to 0.5 degrees Celsius, the processor 1020 may reduce an off-duty of the voltage pulse applied to the thermoelectric element 530 by 4% (1705). As the off-duty of the voltage pulse is reduced, the cooling performance of the thermoelectric element 530 may increase.

[0235] The order of identifying the error of the refrigeration cycle device and the opening of the doors is not limited to the above example. Whether an error has occurred in the refrigeration cycle device and whether the door is opened may be identified independently of each other.

[0236] In a case where the temperature of the cooling sink 570 is lower than 0.5 degrees Celsius, the processor 1020 may identify whether the temperature of the cooling sink 570 is less than or equal to -1.5 degrees Celsius (1706). In a case where the temperature of the cooling sink 570 is less than or equal to -1.5 degrees Celsius, the processor 1020 may increase the off-duty of the voltage pulse applied to the thermoelectric element 530 by 2% (1707). As the off-duty of the voltage pulse increases, the cooling performance of the thermoelectric element 530 may decrease. In a case where the temperature of the cooling sink 570 is higher than -1.5 degrees Celsius and lower than 0.5 degrees Celsius, the processor 1020 may maintain the duty ratio of the voltage pulse applied to the thermoelectric element 530 (1708).

[0237] According to an embodiment, a refrigerator may include: a storage compartment; a cooling sink configured to exchange heat with air in the storage compartment; a thermoelectric element configured to contact the cooling sink and absorb heat; a temperature sensor configured to detect a temperature of the cooling sink; a cooling fan configured to supply the air drawn from the storage compartment to the cooling sink; a cooling duct configured to cover the cooling sink and the cooling fan; and a processor configured to operate the cooling fan to evaporate defrost water accom-

modated in the cooling duct due to defrosting of the cooling sink, and to perform anti-freezing control for the defrost water by adjusting a duty ratio of a voltage pulse applied to the thermoelectric element based on the temperature of the cooling sink.

[0238] The processor may be configured to determine the duty ratio of the voltage pulse applied to the thermoelectric element based on the temperature of the cooling sink at defined time intervals to maintain the temperature of the cooling sink within a temperature range which is preset to prevent the defrost water from freezing.

[0239] The processor may be configured to reduce the duty ratio of the voltage pulse based on the temperature of the cooling sink being lower than or equal to a first threshold temperature. The processor may be configured to increase the duty ratio of the voltage pulse based on the temperature of the cooling sink being higher than or equal to a second threshold temperature higher than the first threshold temperature.

[0240] The processor may be configured to maintain the duty ratio of the voltage pulse based on the temperature of the cooling sink being higher than a first threshold temperature and lower than a second threshold temperature. The first threshold temperature may be lower than the second threshold temperature.

[0241] The processor may be configured to stop an operation of the thermoelectric element and the cooling fan to prevent water overflow in the cooling duct based on a door of the refrigerator being opened for a defined threshold time.

[0242] The refrigerator may further include: an evaporator disposed outside the storage compartment and configured to generate cold air supplied to the storage compartment; and a compressor configured to supply a refrigerant to the evaporator. The processor may be configured to stop an operation of the thermoelectric element and the cooling fan based on an operation of the compressor being stopped.

[0243] The refrigerator may further include: an evaporator disposed outside the storage compartment and configured to generate cold air supplied to the storage compartment; and a return duct configured to guide the air in the storage compartment to the evaporator.

[0244] The refrigerator may further include an evaporator fan configured to move the cold air generated in the evaporator to the storage compartment. The processor may be configured to temporarily stop an operation of the cooling fan to allow moisture in the storage compartment to move to the evaporator through the return duct according to an operation of the evaporator fan, based on an elapse of a defined evaporation time of the defrost water after a start of the operation of the cooling fan for evaporating the defrost water.

[0245] The processor may be configured to temporarily stop the anti-freezing control for the defrost water based on a priority operation condition of the thermoelectric element being satisfied.

[0246] According to an embodiment, in a method for controlling a refrigerator including a cooling sink configured to exchange heat with air in a storage compartment, a thermoelectric element configured to contact the cooling sink and absorb heat, a cooling fan configured to supply the air drawn from the storage compartment to the cooling sink, a cooling duct configured to cover the cooling sink and the cooling fan, and a processor, the method may include: performing, by the processor, a defrosting operation for the

cooling sink; controlling, by the processor, an operation of the cooling fan to evaporate defrost water accommodated in the cooling duct due to defrosting of the cooling sink; detecting, by a temperature sensor, a temperature of the cooling sink; and performing, by the processor, anti-freezing control for the defrost water by adjusting a duty ratio of a voltage pulse applied to the thermoelectric element based on the temperature of the cooling sink.

[0247] The performing of the anti-freezing control for the defrost water may include determining the duty ratio of the voltage pulse applied to the thermoelectric element based on the temperature of the cooling sink at defined time intervals to maintain the temperature of the cooling sink within a temperature range which is preset to prevent the defrost water from freezing.

[0248] The performing of the anti-freezing control for the defrost water may include: reducing the duty ratio of the voltage pulse based on the temperature of the cooling sink being lower than or equal to a first threshold temperature; and increasing the duty ratio of the voltage pulse based on the temperature of the cooling sink being higher than or equal to a second threshold temperature higher than the first threshold temperature.

[0249] The performing of the anti-freezing control for the defrost water may include maintaining the duty ratio of the voltage pulse based on the temperature of the cooling sink being higher than a first threshold temperature and lower than a second threshold temperature. The first threshold temperature may be lower than the second threshold temperature.

[0250] The method may further include stopping an operation of the thermoelectric element and the cooling fan to prevent water overflow in the cooling duct based on a door of the refrigerator being opened for a defined threshold time.

[0251] The refrigerator may include an evaporator configured to generate cold air supplied to the storage compartment, and a compressor configured to supply a refrigerant to the evaporator. The method may further include stopping an operation of the thermoelectric element and the cooling fan based on an operation of the compressor being stopped.

[0252] The refrigerator may include an evaporator configured to generate cold air; an evaporator fan configured to move the cold air generated in the evaporator to the storage compartment; and a return duct configured to guide the air in the storage compartment to the evaporator. The controlling of the operation of the cooling fan may include temporarily stopping an operation of the cooling fan to allow moisture in the storage compartment to move to the evaporator through the return duct according to an operation of the evaporator fan, based on an elapse of a defined evaporation time of the defrost water after a start of the operation of the cooling fan for evaporating the defrost water.

[0253] The method may further include temporarily stopping the anti-freezing control for the defrost water based on a priority operation condition of the thermoelectric element being satisfied.

[0254] According to the disclosure, the refrigerator and the method for controlling the same may improve a cooling efficiency of a storage compartment using a thermoelectric cooler.

[0255] According to the disclosure, the refrigerator and the method for controlling the same may remove defrost water generated by defrosting a cooling sink of a thermo-

electric cooler, thereby preventing a performance of the thermoelectric cooler from deteriorating.

[0256] According to the disclosure, the refrigerator and the method for controlling the same may prevent defrost water accommodated in a cooling duct from freezing.

[0257] According to the disclosure, the refrigerator and the method for controlling the same may prevent water overflow in a cooling duct.

[0258] Meanwhile, the disclosed embodiments may be implemented in the form of a recording medium that stores instructions executable by a computer. The instructions may be stored in the form of program codes, and when executed by a processor, the instructions may create a program module to perform operations of the disclosed embodiments.

[0259] The machine-readable recording medium may be provided in the form of a non-transitory storage medium. Here, when a storage medium is referred to as “non-transitory,” it may be understood that the storage medium is tangible and does not include a signal (electromagnetic waves), but rather that data is semi-permanently or temporarily stored in the storage medium. For example, a “non-transitory storage medium” may include a buffer in which data is temporarily stored.

[0260] The methods according to the various embodiments disclosed herein may be provided in a computer program product. The computer program product may be traded between a seller and a buyer as a product. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or may be distributed through an application store (e.g., Play Store™) online. In the case of online distribution, at least a portion of the computer program product may be stored at least semi-permanently or may be temporarily generated in a storage medium, such as a memory of a server of a manufacturer, a server of an application store, or a relay server.

[0261] Although embodiments of the disclosure have been described with reference to the accompanying drawings, a person having ordinary skill in the art will appreciate that other specific modifications may be easily made without departing from the technical spirit or essential features of the disclosure. Accordingly, the foregoing embodiments should be regarded as illustrative rather than limiting in all aspects.

1. A refrigerator, comprising:
 - a storage compartment;
 - a cooling sink;
 - a thermoelectric element including a cooling layer in contact with the cooling sink, the thermoelectric element configured to, based on a voltage pulse applied to the thermoelectric element, cool the cooling layer and thereby cool the cooling sink;
 - a cooling fan controllable to generate a flow of air from the storage compartment to the cooling sink to be cooled by the cooling sink;
 - a cooling duct covering the cooling sink and the cooling fan, and configured to guide the flow of air from the storage compartment to the cooling sink and to accommodate defrost water generated by water, frozen by the cooling sink, being defrosted;
 - a temperature sensor configured to detect a temperature of the cooling sink; and
 - a processor configured to:
 - control the cooling fan to evaporate the defrost water accommodated in the cooling duct,

perform an anti-freezing control so that the defrost water accommodated in the cooling duct does not freeze, the anti-freezing control including adjusting a duty ratio of the voltage pulse based on the temperature of the cooling sink detected by the temperature sensor.

2. The refrigerator of claim 1, wherein the processor is further configured to determine the duty ratio based on the temperature of the cooling sink detected by the temperature sensor over a time interval so that the temperature of the cooling sink detected by the temperature sensor is maintained within a temperature range which prevents the defrost water accommodated in the cooling duct from freezing.

3. The refrigerator of claim 1, wherein

the processor is further configured to:

reduce the duty ratio based on the temperature of the cooling sink detected by the temperature sensor being lower than or equal to a first threshold temperature; and

increase the duty ratio based on the temperature of the cooling sink detected by the temperature sensor being greater than or equal to a second threshold temperature that is greater than the first threshold temperature.

4. The refrigerator of claim 1, wherein

the processor is further configured to:

maintain the duty ratio based on the temperature of the cooling sink detected by the temperature sensor being greater than a first threshold temperature and lower than a second threshold temperature, and

the first threshold temperature is lower than the second threshold temperature.

5. The refrigerator of claim 1, wherein the processor is further configured to, based on a door of the refrigerator being opened for a threshold time, not apply the voltage pulse to the thermoelectric element and control the cooling fan to not generate the flow of air so that the defrost water accommodated in the cooling duct does not overflow the cooling duct.

6. The refrigerator of claim 1, further comprising:

an evaporator outside the storage compartment and configured to cool air to be supplied to the storage compartment; and

a compressor configured to supply a refrigerant to the evaporator,

wherein the processor is further configured to, based on refrigerant not being supplied to the evaporator by the compressor, not apply the voltage pulse to the thermoelectric element and control the cooling fan to not generate the flow of air.

7. The refrigerator of claim 1, further comprising:

an evaporator outside the storage compartment and configured to cool air to be supplied to the storage compartment; and

a return duct configured to guide air to be cooled from the storage compartment to the evaporator.

8. The refrigerator of claim 7, further comprising:

an evaporator fan configured to move air cooled by the evaporator to the storage compartment,

wherein the processor is further configured to temporarily control the cooling fan to not generate the flow of air so that moisture in the storage compartment is allowed to move to the evaporator through the return duct by operation of the evaporator fan, based on elapse of an

evaporation time for evaporating the defrost water after the cooling fan is controlled to generate the flow of air.

9. The refrigerator of claim 1, wherein the processor is further configured to temporarily stop the anti-freezing control based on a priority operation condition of the thermoelectric element being satisfied.

10. A method for controlling a refrigerator including a storage compartment, a cooling sink, a thermoelectric element including a cooling layer in contact with the cooling sink, the thermoelectric element configured to, based on a voltage pulse applied to the thermoelectric element, cool the cooling layer and thereby cool the cooling sink, the refrigerator further including a cooling fan controllable to generate a flow of air from the storage compartment to the cooling sink to be cooled by the cooling sink, a cooling duct covering the cooling sink and the cooling fan, and configured to guide the flow of air from the storage compartment to the cooling sink and to accommodate defrost water generated by water, frozen by the cooling sink, being defrosted, a temperature sensor configured to detect a temperature of the cooling sink, and a processor, the method comprising:

by the processor,

controlling the cooling fan to evaporate the defrost water accommodated in the cooling duct;

performing an anti-freezing control so that the defrost water accommodated in the cooling duct does not freeze, the anti-freezing control including adjusting a duty ratio of a voltage pulse based on the temperature of the cooling sink detected by the temperature sensor.

11. The method of claim 10, wherein the performing of the anti-freezing control further includes, by the processor, determining the duty ratio based on the temperature of the cooling sink detected by the temperature sensor over a time interval so that the temperature of the cooling sink detected by the temperature sensor is maintained within a temperature range which prevents the defrost water accommodated in the cooling duct from freezing.

12. The method of claim 10, wherein the performing of the anti-freezing control further includes:

reducing the duty ratio based on the temperature of the cooling sink detected by the temperature sensor being lower than or equal to a first threshold temperature, and

increasing the duty ratio based on the temperature of the cooling sink detected by the temperature sensor being greater than or equal to a second threshold temperature that is greater than the first threshold temperature.

13. The method of claim 10, wherein the performing of the anti-freezing control further includes:

maintaining the duty ratio based on the temperature of the cooling sink detected by the temperature sensor being greater than a first threshold temperature and lower than a second threshold temperature, and the first threshold temperature is lower than the second threshold temperature.

14. The method of claim 10, further comprising:

by the processor, based on a door of the refrigerator being opened for a threshold time, not applying the voltage pulse to the thermoelectric element and controlling the cooling fan to not generate the flow of air so that the defrost water accommodated in the cooling duct does not overflow the cooling duct.

15. The method of claim 10, wherein

the refrigerator includes:

an evaporator outside the storage compartment and configured to cool air to be supplied to the storage compartment, and

a compressor configured to supply a refrigerant to the evaporator, and

the method further includes, by the processor, based on refrigerant not being supplied to the evaporator by the compressor, not applying the voltage pulse to the thermoelectric element, and controlling the cooling fan to not generate the flow of air.

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